Copernicus Marine Service

QUALITY INFORMATION DOCUMENT

Global High Resolution Production Centre
GLOBAL_MULTIYEAR_WAV_001_032

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CHANGE RECORD

When the quality of the products changes, the QuID is updated and a row is added to this table. The third column specifies which sections or sub-sections have been updated. The fourth column should mention the version of the product to which the change applies.

Issue	Date	§	Description of Change	Author	Validated By
1.0	August 2019	All	Creation of the document	Stéphane LAW-CHUNE	Mercator Ocean
1.2	March 2020	All	Quality improvement with V1.1	Stéphane LAW-CHUNE	Mercator Ocean
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I EXECUTIVE SUMMARY

I.1 Products covered by this document

The product assessed in this document is referenced as **GLOBAL_MULTIYEAR_WAV_001_032** for the global wave reanalysis describing past sea states over the years 1993-2022 for initial production, then in interim mode for subsequent periods. Recently, time series have been extended backwards in time to 1st January 1980. This product also bears the name of **WAVERYS** within the Global High Resolution (GLO-HR) MFC, corresponding to other global multi-year products like GLORYS, BIORYS, etc. The core of WAVERYS is based on the MFWAM model (Aouf & Lefèvre, 2015), a third-generation wave model that calculates the wave spectrum, i.e., the distribution of sea state energy in frequency and direction on a 1/5° irregular grid. The wave parameters derived from the wave spectrum, such as significant wave height (SWH) or the mean wave period, are delivered globally on a regular grid size of 1/5° with a 3-hour time step. The wave spectrum is discretized into 30 frequencies obtained from a geometric sequence of first member 0.035 Hz and a common ratio of 7.5. WAVERYS considers oceanic currents from the GLORYS12 physical ocean reanalysis (Jean-Michel et al., 2021) and assimilates significant wave height observed from historical altimetry missions and directional wave spectra from Sentinel 1 Synthetic Aperture Radar (SAR) from 2016 onwards. For the 2020-time extension and onwards, wave spectra from the SWIM (Surface Wave Investigation and Monitoring instrument)-CFO sat data are assimilated.

WAVERYS is described in Law-Chune et al. (2021) which also covers some of the results shown in this document.

I.2 Summary of the results

The quality of the WAVERYS wave reanalysis was estimated by: (i) comparing SWH model predictions to HY-2A satellite altimetry data (non-assimilated, open ocean, validation period of 2014-2018); (ii) comparing the mean wave parameters with Copernicus Marine in-situ buoy measurements (coastal data, validation period 1994-2015); and (iii) comparing the dynamics of these mean parameters with respect to ERA5 wave reanalysis for the entire period for which ERA5 data is available. The quality assessment covers a subset of key wave parameters; results of which are summarized below.

Significant wave height (SWH): For the open sea, WAVERYS performs with a global scatter index (SI) of 8.5% with relatively low biases of about 2 cm. The quality of the analysis is better for tropical latitudes than for high latitudes where SI exceeds 10%. This is particularly the case for coastal areas and/or those subject to complex oceanic processes (e.g., current instabilities, tides, etc.), where SI can reach 20%. Generally, the wave model performs worse in light wind conditions (typically summer) than in strong wind conditions that effectively constrain the model (typically winter), with SI differing by a few percent.

Validation of the following quantities is only available for a limited set of coastal moorings located mainly in the Atlantic and Pacific oceans of the Northern Hemisphere:

Peak wave period: Despite the known challenges posed by coastal areas to wave models, the bias in peak periods are only about 10 milliseconds (ms). In an area along the Iberian coast and in the Azores, SI reaches around 40% (compared to the average of 26% over all measurements).

Zero up-crossing period: Mean wave period bias is about 40 ms and SI remains low at about 15%. WAVERYS is particularly efficient at capturing Pacific Ocean swells (scatter index of 11% near Hawaii).

Wave mean direction: The average direction deviation (bias) for all buoy measurements is about 5° while Root Mean Square Difference (RMSD) is 38°. Smaller differences (around 1°) are seen in the Tropics, where swell regimes prevail. Due to wave kinematics, the directions are better constrained near coasts than in the open ocean.

Important notice:

1. **Large and erroneous ERA5 winds for a few dates.**

   The ERA5 winds used for forcing contain erroneous artifacts that are seen by WAVERYS as very strong wind pulses over a short period of time (capped to 50 m/s in WAVERYS). Annex 3 deals with these cases which only concern the Black Sea and Papua New Guinea on a few dates.

2. ***Source code upgrade in 2020 (ECWAM cy38 –> ECWAM cy42)***

   WAVERYS source code is based on EC-WAM (ECWMF). In 2020, this source code was updated to keep pace with developments in Météo-France's real-time system. This update was motivated by the addition of routines enabling the assimilation of SAR spectra from SWIM/CFOSAT. Météo-France documented the transition from cy38 to cy42 by performing a validation in March 2017 with HY2-a altimeter (see Table 1). Cy42 treats bathymetry and sea ice differently, causing differences with Cy38. In any case, these are positive for performance, with a slight reduction of bias and RMSD.

Table 1: Significant wave height validation of cy38 vs cy41 for March 2017 with the HY-2A altimeter. Results are averaged over 50 km squared bins, over more than 153 000 satellite observations.

	Cy38 (REF)	Cy41 (NEW)
Bias (m)	-0.11	-0.10
RMSD (m)	0.30	0.29
SI (%)	9.7 %	9.4 %

I.3 Estimated Accuracy Numbers

This section summarizes system performance in the form of a quick-reference table.

Table 2: Estimated Accuracy Numbers table for WAVERYS essential features.

Variable	Units	Metrics	Values
Open-Ocean Significant wave height	cm, cm, %	mean bias / RMSD / SI	4 / 22 / 8.82
Coastal Significant wave height	cm, cm, %	mean bias / RMSD / SI	-5 / 34 / 20.7
Coastal Peak wave period	s, s, %	mean bias / RMSD / SI	0.36 / 2.3 / 26.2
Coastal Mean wave period	s, s, %	mean bias / RMSD / SI	-0.31 / 0.86 / 14.8
Coastal Mean wave direction	Degree, deg	mean bias / RMSD	4.98 / 37.8

II PRODUCTION SYSTEM DESCRIPTION

Production centre names: Mercator Ocean International and Météo-France

Production system name: GLOBAL_MULTIYEAR_WAV_001_032 (or WAVERYS for short-name)

The CMEMS global wave reanalysis, WAVERYS, was built from the real-time wave system GLO-WAVE-001-027. It therefore contains a similar wave physics and model configuration. The core of the system is based on the wave model MFWAM, a third-generation wave model operated at Météo-France (Aouf et al., 2021).

MFWAM uses the computing code of ECWAM-IFS-42R2 with a dissipation term developed by Ardhuin et al. (2010) (known as “ST4 physics”) added with some improvements obtained from the European research project “My Wave”. To be consistent with the latest version of the CMEMS global wave system, the dissipation source term was also adjusted to reduce the surface stress at the air/sea interface for high and intermediate winds (Rainaud et al., 2019). Typically, with these models, predicting the high frequency part of the spectrum is tricky because these are the frequencies where wind energy is transferred to feed the waves, but also the part of the spectrum subject to the most dissipative processes. WAVERYS corrects the upper part of the spectrum (high frequencies) to a Phillips tail (theoretical spectrum) to ensure realistic spectral content.

WAVERYS’s bathymetry is generated by using 2-minute gridded global topography data from ETOPO2/NOAA. The native model grid is irregular with latitudinal distances decreasing poleward. At the equator, the distance in the latitudinal direction is fixed with a grid size of 1/5°. The wave spectrum is discretized into 24 directions and 30 frequencies starting from 0.035 Hz and up to 0.58 Hz. A minimum water depth is set to 5 m, which is more than sufficient at the 1/5° global resolution to represent shoaling effects.

Generally, the quality of wave forecasts is severely constrained by surface winds, and the WAVERYS wave reanalysis is driven by winds and ice-cover from the latest climate reanalysis produced at ECMWF, namely ERA5 (Hersbach et al., 2020). ERA5 climate reanalysis has an average resolution of 31 km, and although hourly analysis fields are available, three hourly winds are used in WAVERYS for consistency with the other forcing data such as oceanic currents and data-assimilation. ERA5 reanalysis has significantly improved the ERA-interim (ERAI) reanalysis. Appendix 1 provides an illustration of the differences in results in terms of wind-sea modelled during a cyclonic event using ERAI and ERA5.

Producing a reanalysis involves using past observations of model variables to have the best possible initial states for the forecast segments. Due to their large spatial coverage and high temporal repetitiveness, significant wave height observations from satellite are crucial for operational models. WAVERYS takes advantage of most historical altimetry missions. These data were reprocessed (corrected and validated) by CERSAT-IFREMER. The assimilation of SWH is performed over the entire period with a time step and an assimilation window of 3 hours. The assimilation system can use conjointly SWH and directional wave spectra provided by measurements from satellites. The assimilation of wave spectra consists in performing optimal interpolation on mean energy and wavenumber components of dominant wave trains called partitions. Since March 2016, the level 2 Synthetic Aperture Radar (SAR) directional wave spectra from Sentinel-1A are provided by the European Space Agency and are also assimilated to correct the low frequency part of the wave spectrum which controls the swell partitions. WAVERYS is the first available global wave reanalysis that assimilates SAR wave spectra measured from space. Since February 2020, SAR wave spectra from CFO-SAT are also assimilated, also obtained for ESA CCI Sea State project (Dodet et al., 2020). The assimilation of these

data makes it possible to control long swells very effectively; typically those generated in the Southern Ocean (Aouf et al., 2021).

Table 3: List of assimilated satellite data and their period of use.

Satellite	Period of use
ERS1	01/01/1993 - 02/06/1996
TOPEX/POSEIDON	01/01/1993 - 08/10/2005
ERS2	15/05/1995 - 04/07/2011
GFO	07/01/2000 - 01/08/2006
Jason1	15/01/2002 - 21/06/2013
Envisat	14/05/2002 - 08/04/2012
Jason2 et Jason 3	04/07/2008 - present
Cryosat	14/07/2010 -present
Saral	01/03/2016 - present
Sentinel 3A	01/03/2016 - present
SAR CFOSAT/SWIM	01/03/2016 - present
SAR CFOSAT/SWIM	01/02/2020 - present

Ocean currents can strongly impact wave propagation and induce a change in wave frequency, energy and direction (Philipps, 1984). To provide the most accurate sea state, WAVERYS considers 3-hourly surface currents provided by the global physical reanalysis of CMEMS called GLORYS12V1(GLOBAL_REANALYSIS_PHY_001_030 CMEMS product). GLORYS12 currents are interpolated on the native grid of the wave model at 1/5°. This approach has shown very positive results for wave fields interacting with high-energy current systems such as the Agulhas.

Note on wave spectra: Wave spectra are what the wave model calculates. They are often used to compute all the average parameters that are distributed in CMEMS (significant height, period, partitions, etc.). These wave spectra are not directly recorded along the model run because they are very heavy in terms of data (24 directions × 30 frequencies × all grid points × output time steps). Wave spectra are then lost during the production, but they are nevertheless recorded at a few sporadic points for validation purposes. These few validation wave spectra are not disseminated within CMEMS.

III VALIDATION FRAMEWORK

To measure the impact of the different forcing elements on the quality of wave forecasts, a series of preliminary tests were carried out during the design of WAVERYS (see Appendix 2). These were intended first to test the impact of ERA5 winds against ERAI winds, then to assess the addition of GLORYS currents and the assimilation system separately. These tests were carried out for the year 2010. The SWH data of the JASON2 altimeter was kept aside for validation purpose. The results can be found in appendix 2, showing that the final solution for WAVERYS allows to obtain scatter index for the global open-sea below 10%; i.e., almost the same order of magnitude as the CMEMS real-time global wave system.

The wave model computes at each point of the domain grid the distribution of sea state energy according to frequencies and propagation directions. This is what is called the local wave spectrum and average wave parameters provided by CMEMS are derived from its shape and spectral moments. They give to end-users practical and relevant information.

For the validation of WAVERYS, the following average wave parameters were considered:

a) Significant wave height (**SWH**) is probably the most important parameter because it corresponds to the estimate of wave height that would be made by a visual observer at sea. Physically, it represents the mean value of highest one-third of wave heights. Its validation tests the wave model's ability to distribute and disperse wind energy in frequency and direction, if the wind forcing is considered as correct. This quantity is approximated in the wave model from the zeroth moment of the wave spectrum.
b) Peak wave period (**PWP**) represents the period for which most of the energy is found. It is derived from the wave spectrum using the most energetic wave frequency. Its quantity represents the way in which the wave model adjusts the energy across frequencies (balance between growth, dissipation, and wave-wave interactions).
c) Average zero-upcrossing period (**TM10**) represents the time between exceedances of mean water level on the front face of successive waves. This is the wave period classically derived from a temporal wave observation (from wave buoys). It can be approximated by a wave model using the second and zeroth moments of the wave frequency spectrum and represents the average frequency energy distribution.
d) Average wave direction (**VMDR**) represents the average direction where most of the wave energy propagates. It is connected to the model's ability to predict the energy distribution in terms of direction. It is approximated using the first directional moments of the wave spectrum.

The validation of WAVERYS is based on the direct comparison of the above quantities to three relevant data sets.

1. The HY-2A satellite of the Haiyang series for SWH validation from 2013 onwards was used. HY-2A is China's first ocean dynamic environment satellite and as it provides independent data for WAVERYS, it is well suited for validation purposes. This validation focuses on ***the large-scale and offshore ocean***. The HY-2A SWH used in the validation are provided by the French Space Agency (CNES), which reprocessed the data recently (Jiang et al 2018). The Level 2 HY-2A wave data are downloaded from the users' portal AVISO+ : https://www.aviso.altimetry.fr/fr/accueil.html

2. A set of wave buoys from CMEMS *INSITU_GLO_WAVE_REP_OBSERVATIONS_013_045* product were used. This product integrates wave observations aggregated and validated from the Regional EuroGOOS consortium (Arctic-ROOS, BOOS, NOOS, IBI-ROOS, MONGOOS) and Black Sea GOOS as well as from National Data Centres (NODCs) and JCOMM global systems (Argo, GOSUD, OceanSITES, GTSPP, DBCP) and the Global telecommunication system (GTS) used by the Met Offices. This is more ***coastal*** than open-ocean and focuses on the good propagation of wave trains arriving punctually at the position of the buoys.

3. WAVERYS over its entire series with the ERA5 wave reanalysis was compared to ensure its ***inter-annual consistency***. Table 4 provides information on the technical differences between ERA5 and WAVERYS. A similar comparison was made between WAVERYS and the CMEMS near real-time global wave forecasting system.

Table 4: Main differences between WAVERYS and ERA5 wave reanalysis.

	Physics for source terms	*Coupling with atmosphere model*	*Current refraction*	*Horizontal resolution*
ERA5	Jansen 2014	in: air density, 10 m wind and surface roughness out: Charnock parameter	NO	0.5°
WAVERYS	ST4	NO	3h GLORYS12V1 currents	0.2°

Metrics: If $Model(t)$ and $Obs(t)$ are a wave parameter time series from the model and observations (taken at the same time and location) respectively, and $E[.]$ being the temporal mean operator, then the time series are analysed using the following diagnostics:

- Average differences or bias:

$$bias = E[Model(t) - Obs(t)] \qquad (1)$$

- Root mean squared deviation (RMSD: average squared difference which contain the bias of the model and the standard deviation of the model with respect to observations ($S_{Model,Obs}$).

$$RMSD = \sqrt{E[Model(t)^2 - Obs(t)^2]} = bias + S_{Model,Obs} \qquad (2)$$

- Scatter index (SI) or coefficient of variation of RMSD: Normalization of RMSD by the average observed value. The normalized value is less sensitive to extreme values in the target variable.

$$SI = \frac{\sqrt{E[(Model(t) - E(Model(t))^2 - (Obs(t) - E(Obs(t))^2]}}{E[Obs(t)]} \qquad (3)$$

- Correlation coefficient (r): A correlation between variables indicates that as one variable changes in value, the other variable tends to change in a specific direction. A correlation coefficient is a quantitative assessment that measures both the direction and the strength of this tendency to vary together. Only linear relationships are covered by the Pearson correlation coefficient that varies between [-1, 1].

$$r = \frac{COV(Obs,Model)}{S_{Obs}.S_{Model}} \qquad (4)$$

With $COV(.)$ the covariance operator and $S_.$ the standard deviation operator.

IV VALIDATION RESULTS

IV.1 Significant wave height (SWH)

IV.1.1 Large scale and open ocean validation: comparison to HY-2A altimeter

To check the quality of SWH from HY-2A reprocessed by CNES, the data are filtered and compared to reference altimetry mission such as Jason-2. This comparison reveals a strong underestimation of SWH observed by HY-2A (on average -12cm). A linear regression (y=ax+b) is then used to correct this bias and to ensure a HY-2A data quality closer to Jason-2.

The SWH validation of WAVERYS is performed globally on a grid size of 0.5° and a time window of 3hours. The period starts from 01/01/2014 until 26/12/2018.

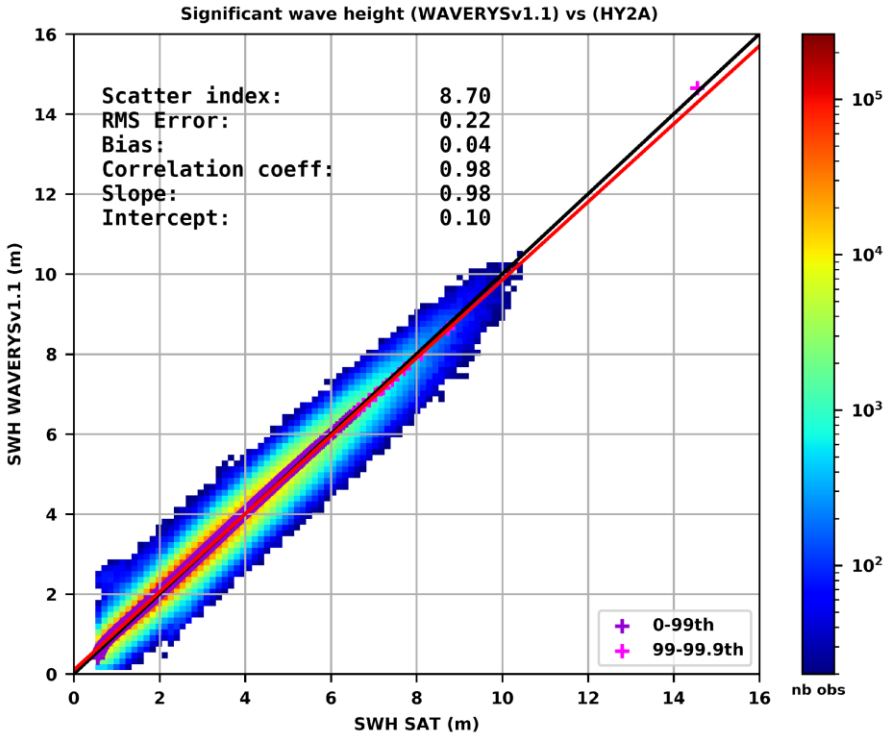


Figure shows the scatter plots of SWH from wave reanalysis (WAVERYS and ERA5) and HY-2A over the considered period with 9,425,957 collected observations. The scatter index for SWH is 8.7% for WAVERYS versus 9.6% for ERA5. The improvement of WAVERYS over ERA5 for these statistics is about 9%. The slope and intercept of the WAVERYS best fit line are 0.98 and 0.1 respectively, thus better than those for ERA5 which are 0.93 and 0.16 respectively. The deviations of ERA5 from observation mainly concern quantiles above the 99th percentile and SWH > 6.5 m.

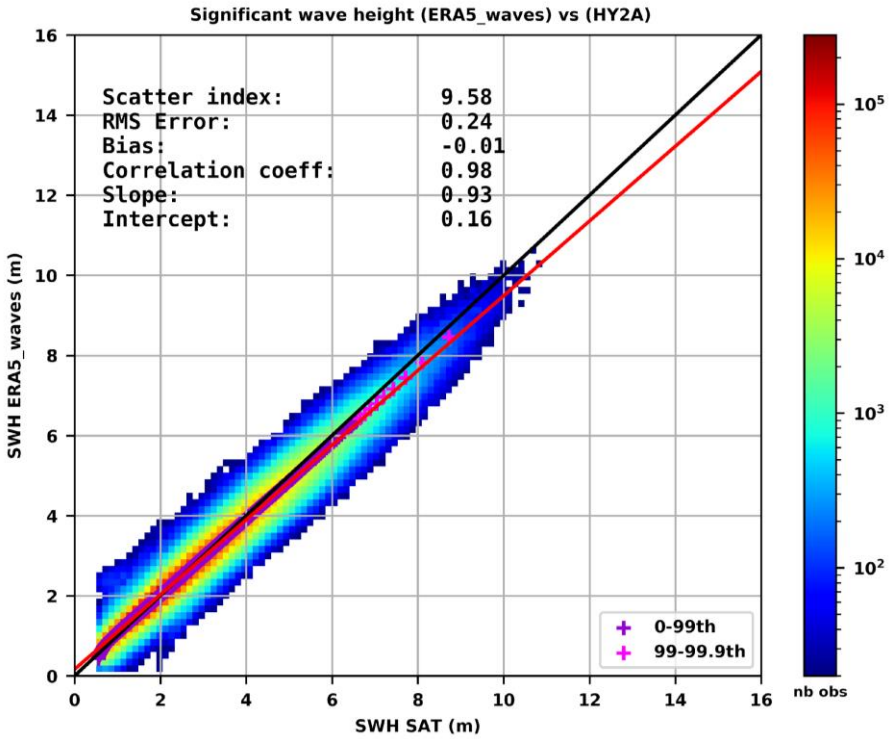


Figure 1. Scatter diagram between HY-2A satellite SWH and the model results (ERA5 on the top, WAVERYS on the bottom). Colours represents the observation density. Statistical scores are given, as well as the slope and intercept of the associated linear regression. Black crosses represent quantiles vs. quantile values, the pink crosses represent the highest values (99-99.9th percentiles) (continues in next page).

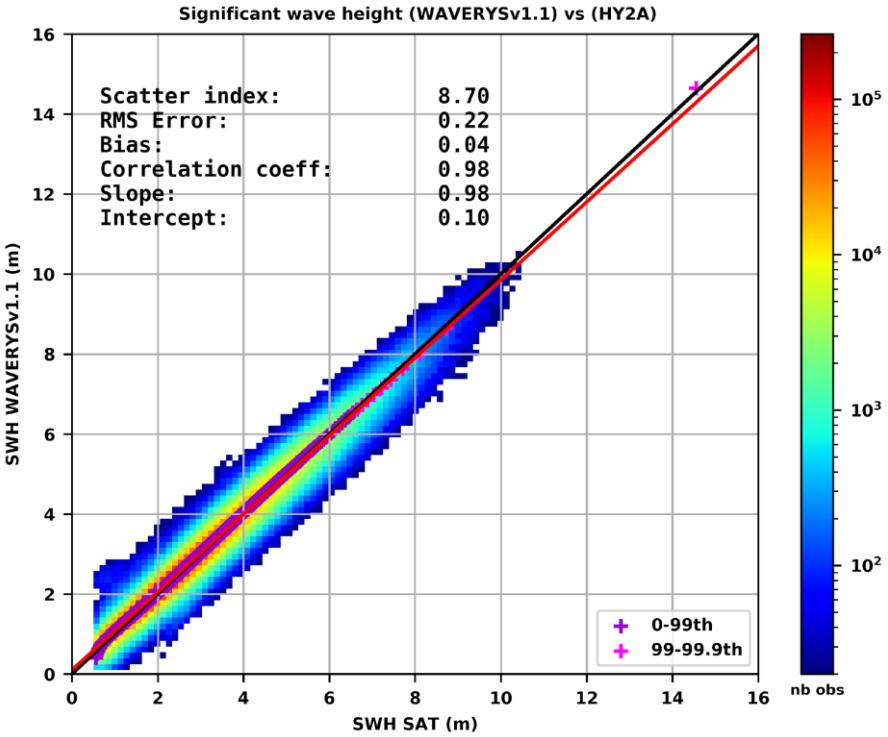


Figure1. (continued) Scatter diagram between HY-2A satellite SWH and the model results (ERA5 on the top, WAVERYS on the bottom). Colours represents the observation density. Statistical scores are given, as well as the slope and intercept of the associated linear regression. Black crosses represent quantiles vs. quantile values, the pink crosses represent the highest values (99-99.9th percentiles).

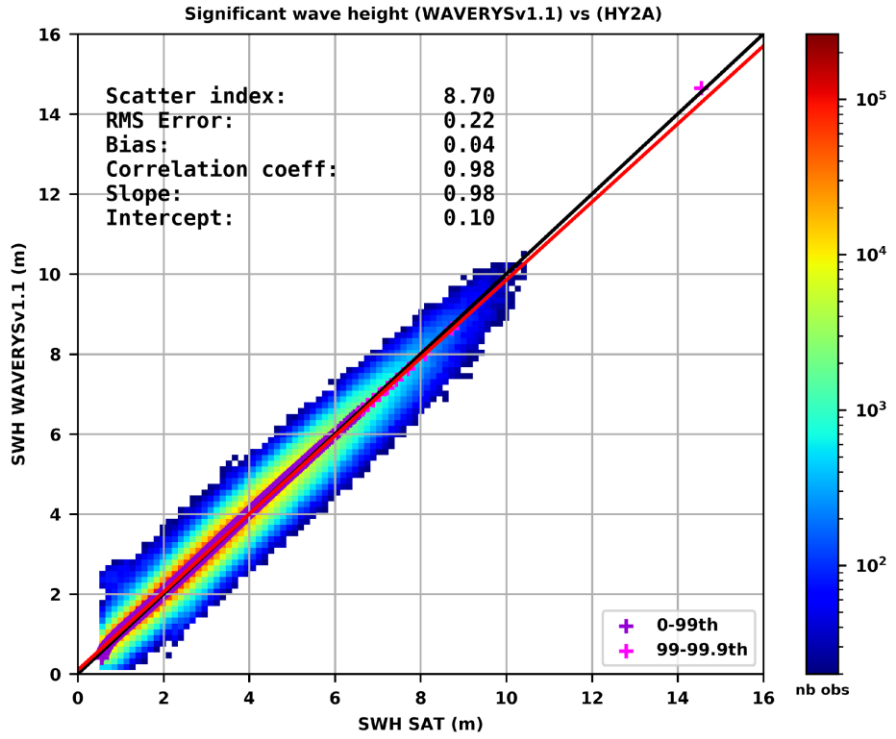


Figure equally presents the average bias and scatter index maps over the global domain. Clearly, the models show the most significant biases in polar regions, where the positive bias reaches nearly 80 cm close to the marginal ice zones due to non-represented interactions with ice in both models. Also, measurement errors should be significant for HY-2A data at these latitudes. Other regions show a systematically low bias, like the enclosed seas and complex coastal areas; likely due to misjudged fetch. Interestingly, the sign of the biases at mid-latitudes and those subject to westerly winds are reversed between WAVERYS and ERA5. These biases are of the order of about 10 cm. WAVERYS SWH bias generally stays below 10 cm, except for southern latitudes > 60°S; likely due to an overestimation of winds in these regions.

The scatter index maps show a very good performance of WAVERYS where the index at mid- and intermediate latitudes is less than 9%. The scatter index increases to a range between 12 and 16% in coastal ocean areas and closed seas. The low scatter index values found in the open ocean for both reanalysis systems cover more domains in WAVERYS than in ERA5. This is quite clear in areas dominated by strong currents, such as the Pacific Equator, the Agulhas Current, Gulfstream, Brazilian, and California currents. The wave physics at mid latitudes in the Atlantic Ocean and Northern Indian Ocean seems also better modelled with WAVERYS.

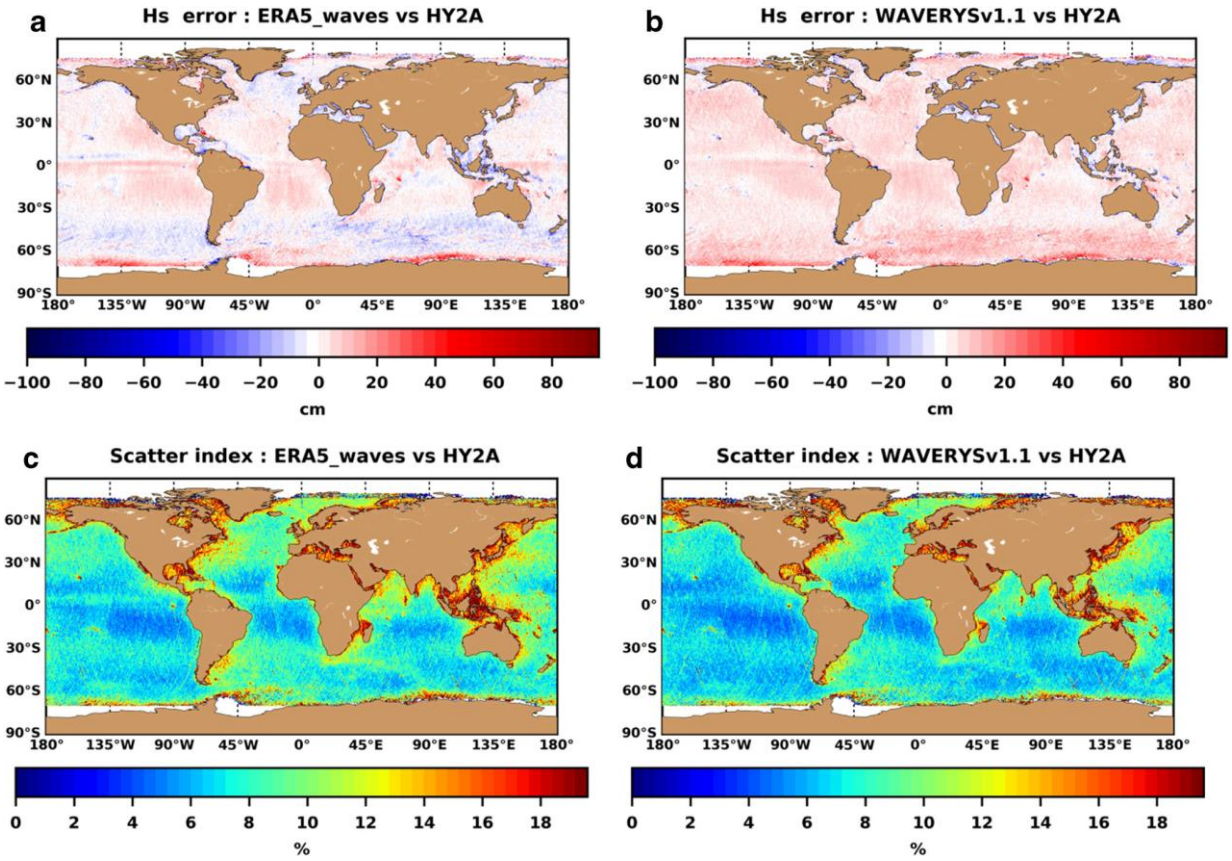
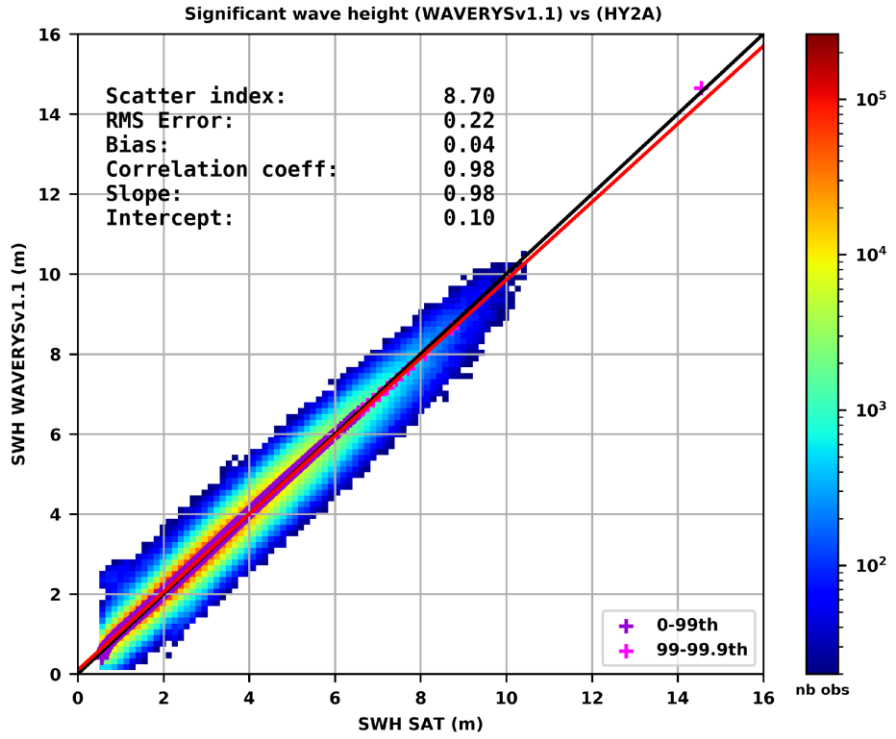


Figure 2. Maps of mean error (top row) and scatter indices (bottom row) of SWH with respect to HY-2A for the period 2013 to 2018; (a) and (c) show ERA5 results while (b) and (d) are those for WAVERYS.

Figure 3 shows the averaged performance metrics of WAVERYS and ERA5 for different ocean regions. The selection of the ocean regions does not consider model points within 25km of coasts (which can produce results different from



Figure, where all observations were considered). The average and global SWH bias for WAVERYS is positive and small (< 4 cm). ERA5 has nearly zero overall bias as positive/negative regional values cancel out. In ocean areas dominated by strong currents, such as the Southern Ocean or the Kuroshio and Agulhas Currents, WAVERYS shows a SWH scatter index that is improved by about 10% compared to ERA5 (due to the inclusion of current forcing from the GLORYS reanalysis). For instance, in the Agulhas Current, this led to a 20% improvement compared to ERA5. In shallow water areas such as in the Iberian Biscay Irish (IBI) domain, WAVERYS also shows a better scatter index for SWH than ERA5. Near the Indonesian archipelago, where models typically struggle to propagate waves due to insufficient resolution, both models give the same results with a scatter index of 18%.

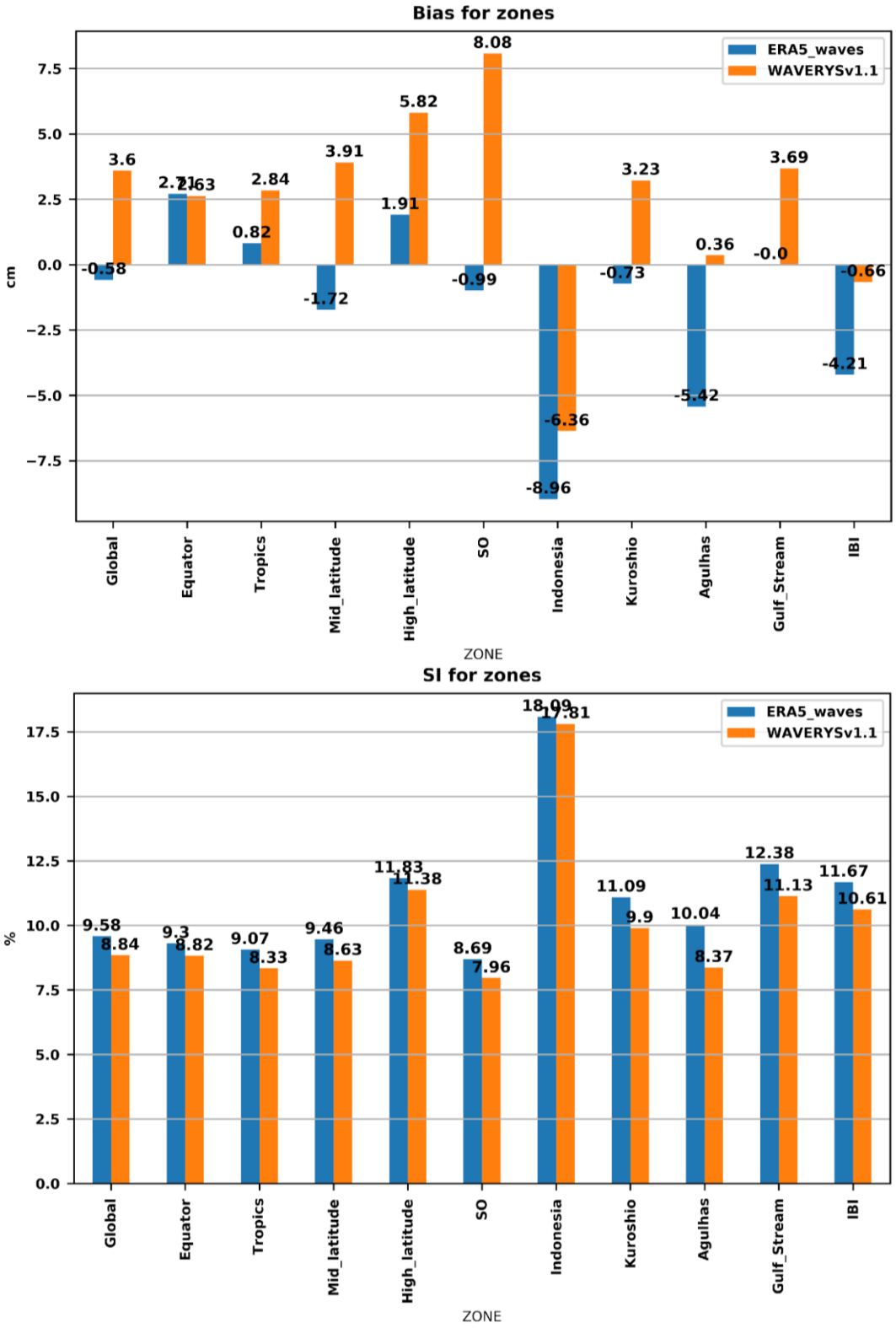


Figure 3. Statistical analysis for different ocean regions for the period from 2014 to 2018. Top and bottom figures show the bias and scatter index of SWH respectively. Blue and red colours represent WAVERYS and ERA5 respectively. Ocean regions are defined according to the following separations: Equator: |lat|<3°; Tropics: |lat|<23.5°; Mid_latitude: 23.5°<|lat|<60°; High_latitude: |lat| >60°; SO: all the Southern-Ocean; IBI: Iberian, Biscay and Irish CMEMS zone.

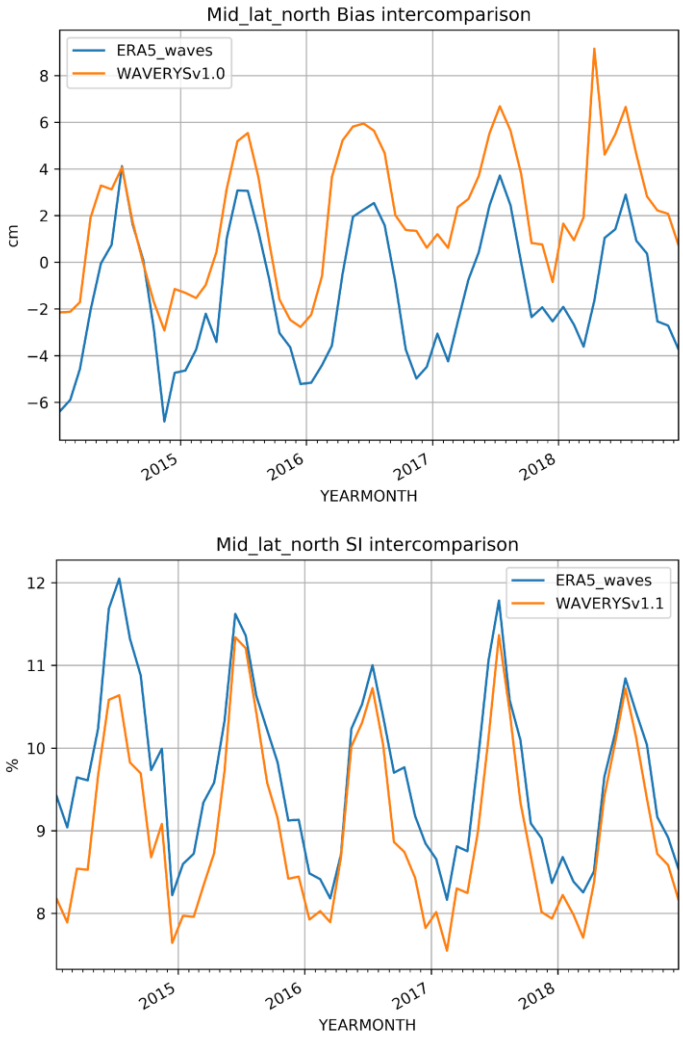


Figure 4. Time series for bias and scatter index of SWH averaged for northern mid-latitude region (23.5°N<lat<66.5°N).

At northern mid-latitudes, SWH bias and scatter index show clear seasonal variations, with larger values in summer (Figure 4). This is likely due to a better response of the model for stronger winds. WAVERYS has an SI of about 8% in winter and about 10.5% in summer and shows a better overall performance than ERA5, in particular in winter. The average bias of WAVERYS is about 1 cm compared to -2.5 cm for ERA5.

IV.1.2 Coastal and in-situ validation: comparison to CMEMS in situ data

In-situ observations by wave buoys are mostly restricted to near-shore locations (Figure 5), which complicates the validation for a wave system like WAVERYS which is designed for deeper, open water settings. Wave buoys are mostly located at mid and high latitudes of the northern hemisphere, and close to the coasts of North America and Europe. This is also where the SWH time series are the longest,

containing data for two to six years, which facilitates the analysis and validation of seasonal cycles. Most tropical data are from the northern hemisphere (Caribbean islands, Hawaii, and offshore India), although Indian Ocean measurements are short and do not cover an entire seasonal cycle. Like Indian buoys, the Italian buoy network only covers a few months (2-6 months). For the remaining buoys, only a few measurement points are available worldwide, e.g., three equatorial buoys (off French Guiana, Colombia, and Kiribati Island), and two buoys respectively off Chile and off Florianopolis, Brazil.

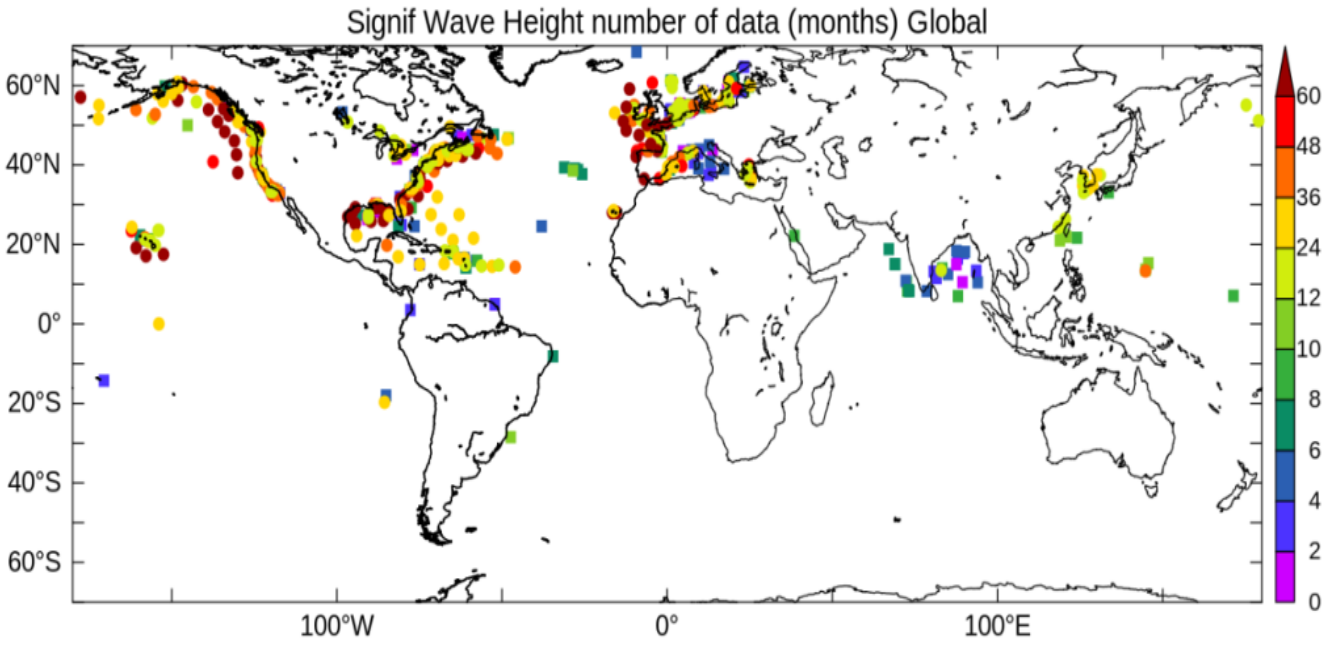


Figure 5. Spatial distribution of CMEMS wave buoys (INSITU_GLO_WAVE_REP_OBSERVATIONS_013_045) and duration of measurements (colours, in months) for SWH. Time series of less than one year are marked by a square instead of a circle.

WAVERYS underestimates the SWH by about 5 to 15 cm on average for most of the buoy locations (Figure 6). The best scores are achieved in tropical latitudes with SWH biases of less than 10 cm. On the other hand, SWH is overestimated in some locations by more than 20 cm, such as in the North Sea, near the Azores and Cape Verde, and in the Bay of Bengal Sea and Arabian Sea. These misfits could be related to some local bias in the wind forcing or some local issues in wave dissipation budget (like shoaling effect or current-induced wave breaking).

RMSD of WAVERYS with respect to observations is less than 20 cm for tropical latitudes in the Atlantic and Pacific oceans (Figure 7). However, RMSD increases to more than 50 cm near the Azores, Cap Verde, Saint Pierre et Miquelon, and Taiwan. For the mid-latitudes and their strong variability in winds and surface currents, RMSD obtained from WAVERYS is in the range of 35 cm.

The best SI values of about 10 are obtained for the eastern tropical Atlantic and the tropical Pacific (around the Hawaii area) (Figure 8). This indicates for instance, that swell propagating from the Southern Ocean is well represented by WAVERYS. SI ranges between 10 and 20% for the US and western European coasts. For specific coastal areas such as Hudson Bay, the Gulf of Alaska, and some locations around Italy, the scatter index increases strongly to more than 30%. This is expected because of the grid size in WAVERYS.

Correlations are very high, above 0.9 for most buoy locations (Figure 9). This confirms that WAVERYS and observations are capturing the same physical processes in the ocean areas (swell propagation, change induced by surface currents for example). Only rarely, do buoys exhibit low correlations of about 0.4 (e.g., near French Guiana, Cap Verde, or in the norther part of the Black Sea). This can be explained by the strong uncertainties related to buoy data in such coastal areas, well known for moving sand bars and where the bathymetry is not accurate because of its variability.

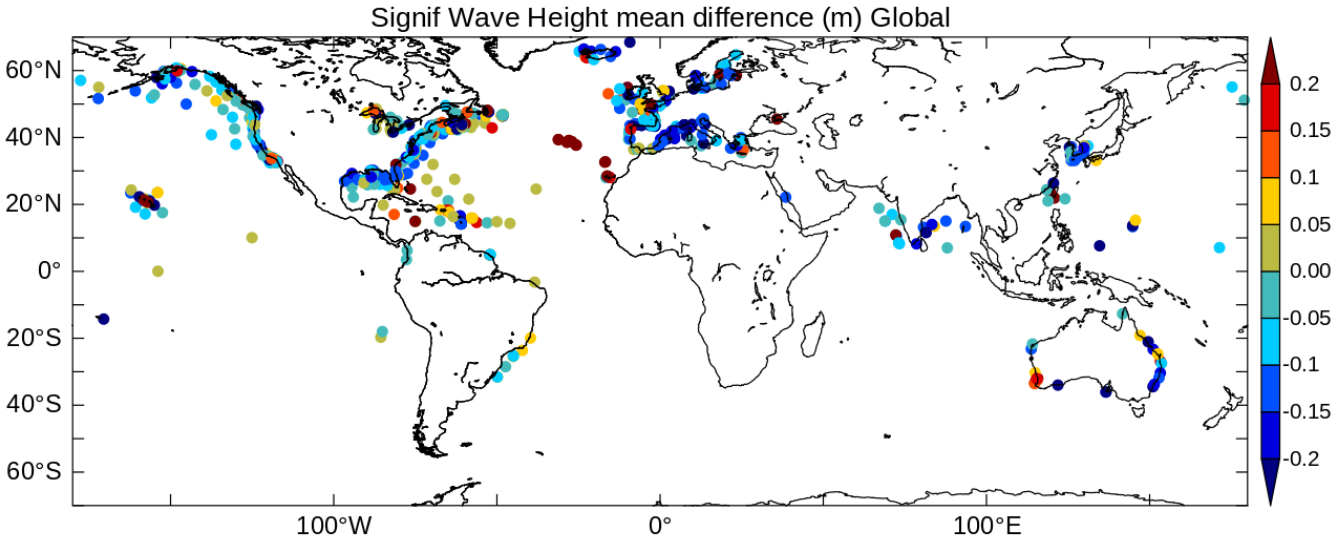


Figure 6. Mean difference of SWH in meters between WAVERYS and CMEMS buoys averaged over the observed period.

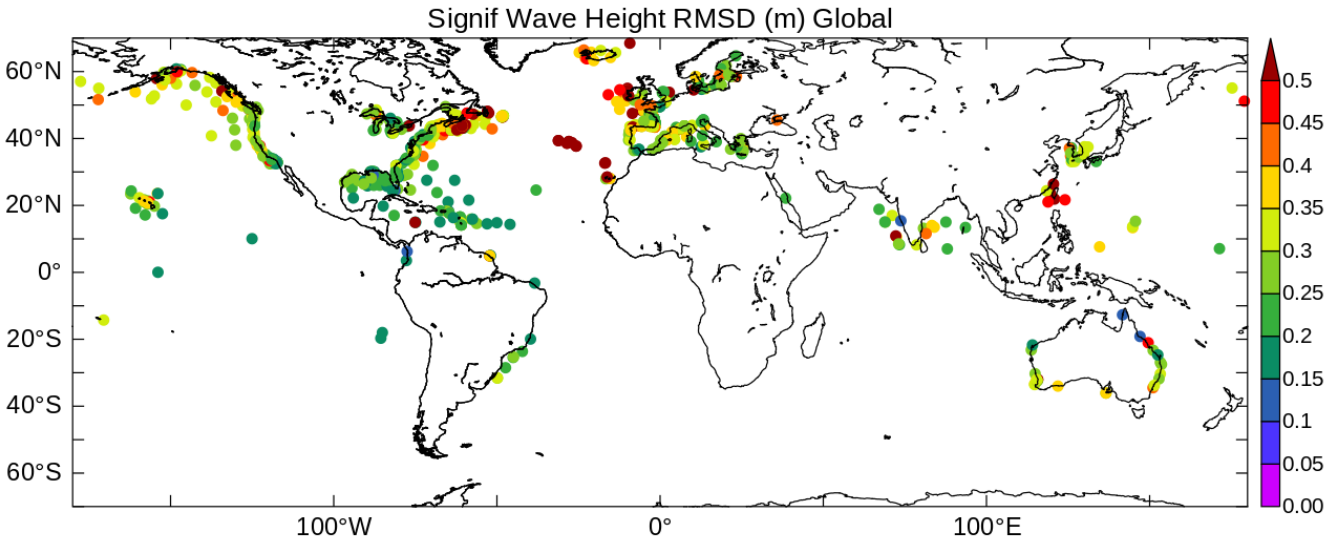


Figure 7. RMSD of SWH in meters for WAVERYS with respect to CMEMS wave buoys.

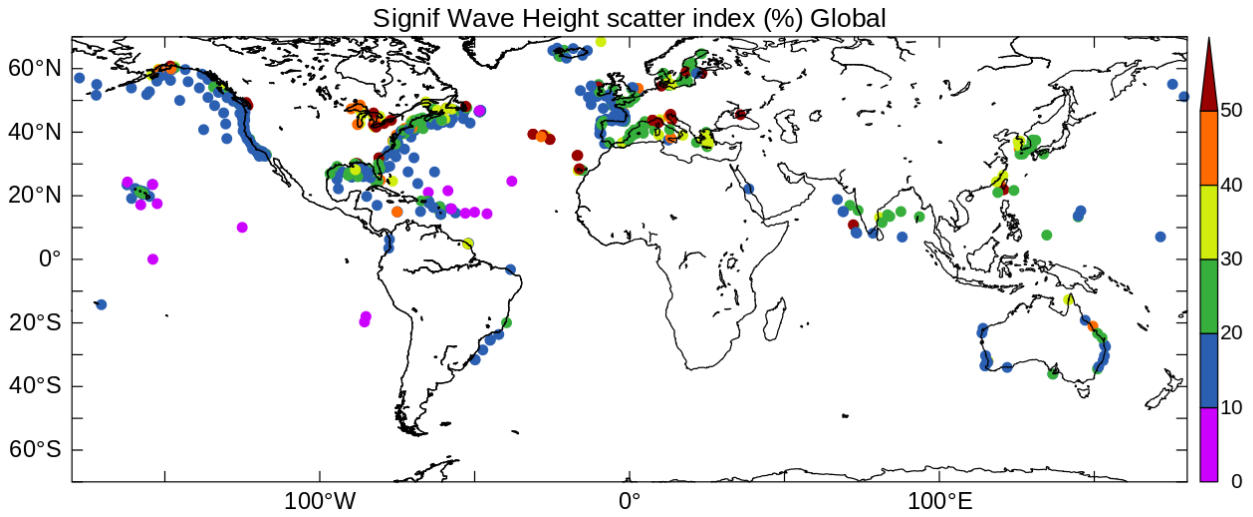


Figure 8. Scatter index of SWH (in %) for WAVERYS with respect to CMEMS wave buoys.

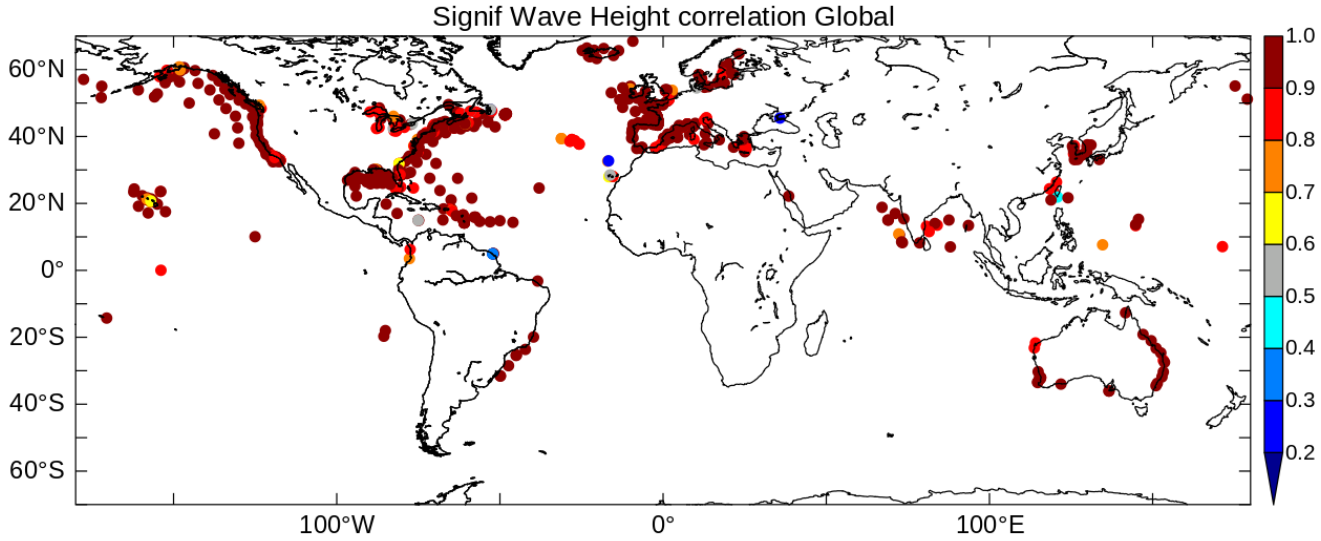


Figure 9. Correlation coefficient of SWH for WAVERYS and CMEMS wave buoys.

summarizes the statistical parameters obtained in different ocean areas. On global average, WAVERYS SI is 20.7%, bias is 6 cm, and RMSD is 34 cm. In tropical areas (Hawaii and Caribbean Sea), SI is satisfactory with 14.5%. The European Atlantic coasts as well as North America's West coast show SI values of around 18%. Areas with SI > 35 % are the Western Pacific (with a low bias of 3 cm) and the North Sea (with a bias of 5 cm).

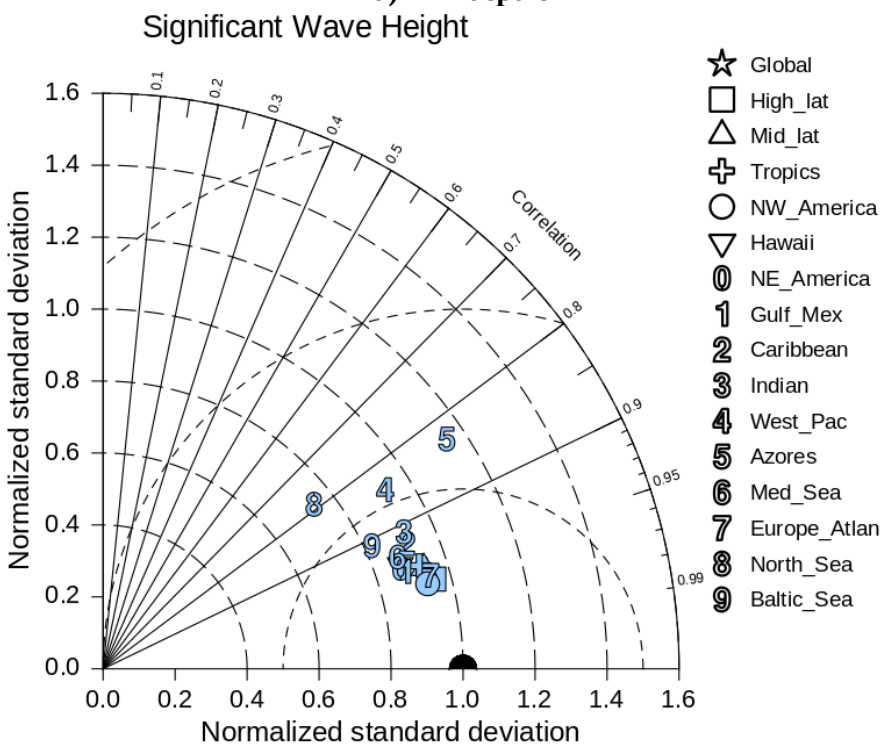
Table 5. Statistical analysis of significant wave height (m) for different regions with respect to in-situ buoys. Period: 1994-2015.

Region	Difference (m)	Scatter Index (%)	RMSD (m)	Number of data values
Global	-0.05	20.7	0.34	13 670 206
High latitudes	-0.07	19.8	0.40	1 995 545
Intermediate latitudes	-0.05	21.4	0.33	10 693 138
Tropics	-0.02	14.5	0.24	981 523
North America West coast	-0.07	15.8	0.32	3 300 860
Hawaii	-0.04	12.8	0.27	428 239
North America East coast	-0.06	23.0	0.31	1 964 820
Gulf of Mexico	-0.09	22.2	0.23	662 768
Caribbean Sea	-0.01	14.5	0.23	426 880
Indian Ocean	0.00	26.4	0.40	42 513
West Pacific	-0.05	41.0	0.45	231 578
Azores	0.52	47.1	0.80	61 992
Med Sea	-0.12	30.4	0.30	11 25 516
Europe Atlantic coast	-0.04	19.0	0.37	2 290 047
North Sea	-0.12	42.6	0.57	125 188
Baltic Sea	-0.11	35.2	0.33	339 577

Taylor diagrams (

) showing shelf measurements separated from open sea measurements (depths > 200 m) confirm that the model responds better offshore than in coastal areas as all graphical representations of offshore measurement points are closer to unity (black points). This is very clear for the Azores measurement for example, which exhibit a correlation of 0.75 near the coast and 0.85 further offshore.

a) All depths



b) Deep water

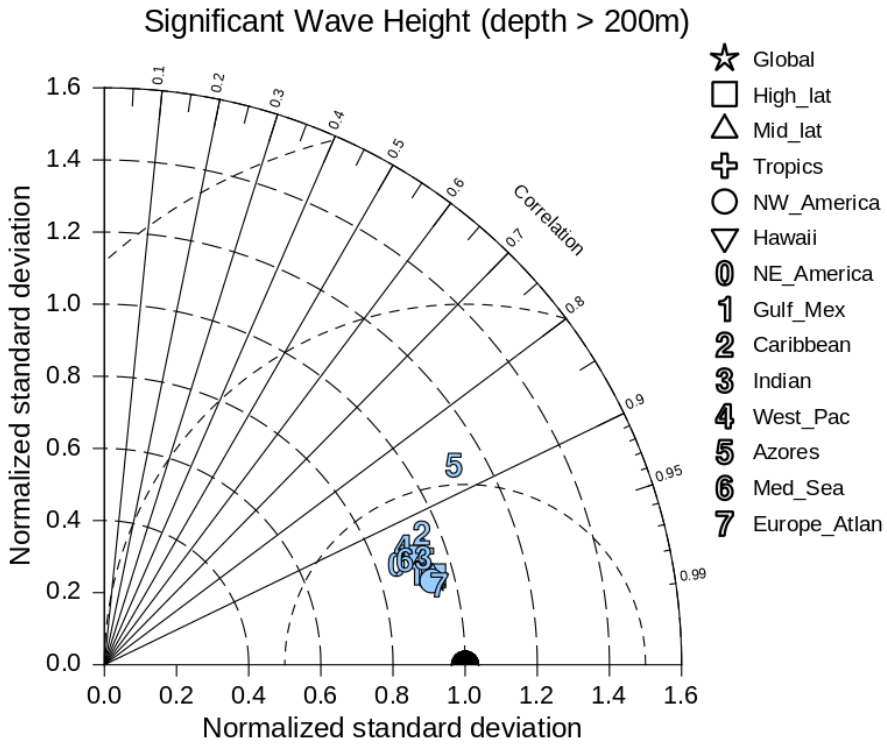


Figure 10. Normalized Taylor diagrams for SWH of WAVERYS compared to CMEMS buoys. Statistics for different regions, with different depth integrations : a) all depths considered, b) only waters with depth > 200 m, c) only waters with depth < 200 m (continues in next page).

a) Shallow water

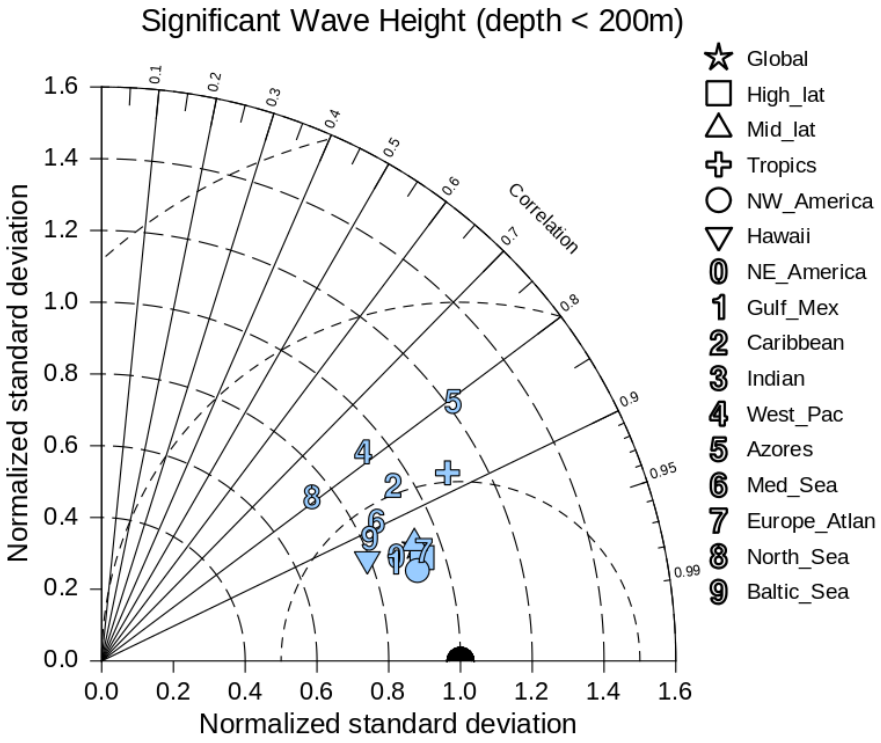


Figure 10. (cont.) Normalized Taylor diagrams for SWH of WAVERYS compared to CMEMS buoys. Statistics for different regions, with different depth integrations : a) all depths considered, b) only waters with depth > 200 m, c) only waters with depth < 200 m

IV.1.3 Inter-annual consistency: comparison to ERA5 wave climatology

In this section, time series of SWH of WAVERYS is compared with those provided by ERA5 reanalysis. To this end, WAVERYS data have been interpolated to ERA5 grid size and the difference is computed between the two systems for each 3h timestep. Time series of monthly averages of difference, standard deviation, and scatter index have been computed (Figure 11).

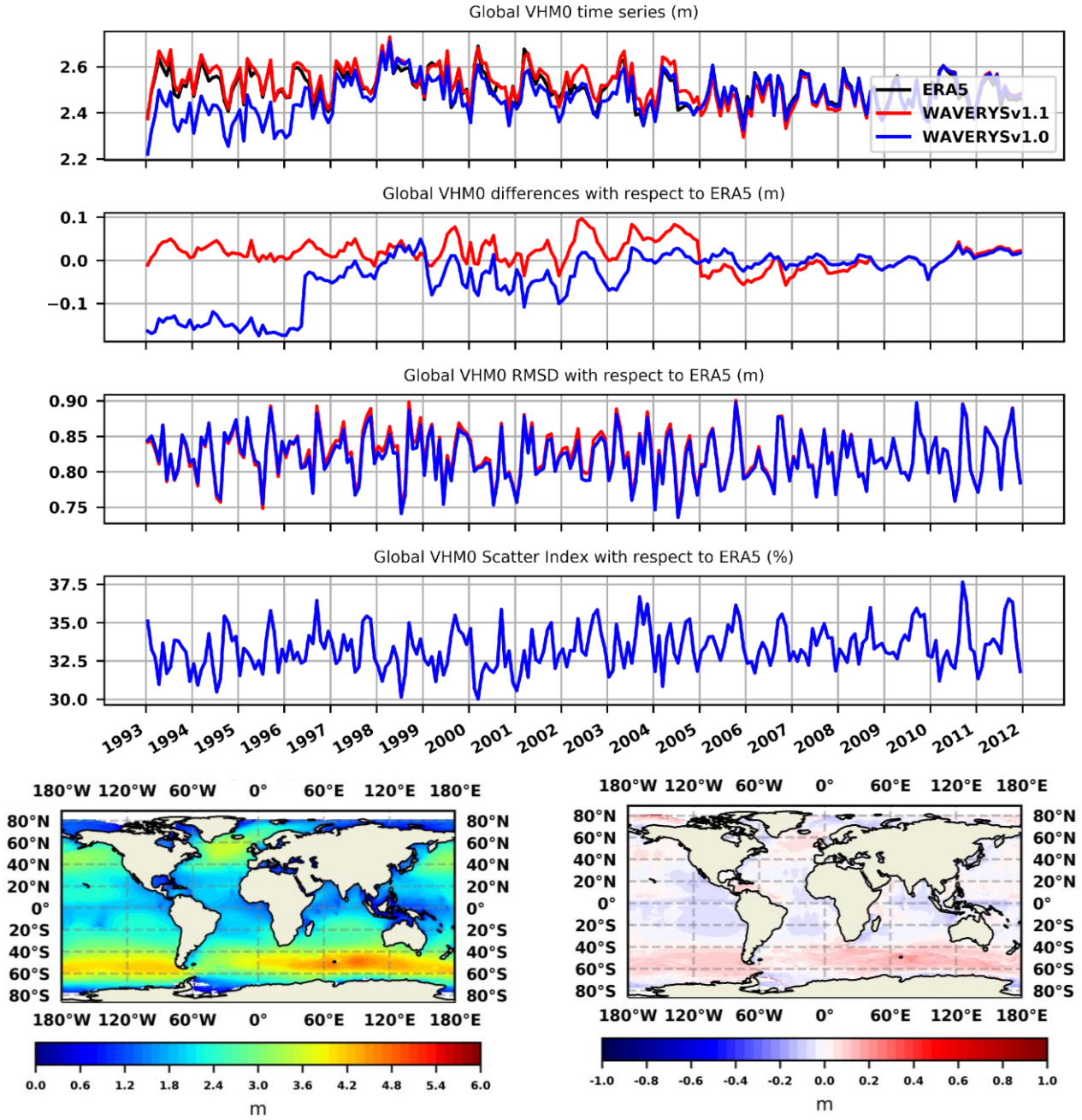


Figure 11. Comparison of the SWH time series of WAVERYS and ERA5 (top panels). V1 corresponds to the first release of WAVERYS which has high biases (termed "differences" in the figure panel) during the first altimetry missions (ERS1 and TOPEX). Those biases are corrected in V1.1 (bottom row). Left panel: mean SWH (m) values of WAVERYS for the whole period; Right panel: mean differences of SWH (m) between WAVERYS and ERA5 over 1993-2012.

WAVERYS and ERA5 SWH times series behave consistently for the total period with a slight overestimation of 5 cm until 2005, followed by an underestimate of 5 cm for the rest of the series. Over the whole period, RMSD between WAVERYS and ERA5 is around 80 cm, and SI around 32% for WAVERYS with respect to ERA5. The bottom-right panel in Figure 11 shows that WAVERYS has higher SWH by about 20 cm with respect to ERA5 for the most energetic areas such as depression rails in mid-latitudes, or near the coast (Indonesian continent for example). In contrast, WAVERYS open ocean wave trains are less energetic than ERA5 by about 15 cm.

IV.2 Peak wave period (PWP)

IV.1.4 Coastal and in-situ validation: comparison to CMEMS in situ data

Figure 12 shows the few CMEMS wave buoys that provide wave period measurements. Unfortunately, Chinese and Indian buoys used in Section IV.1 do not monitor periodic aspects of waves. In addition, wave period measurements are spatially sparser and shorter in duration.

Figure 13 shows the average differences (in seconds) between the wave periods modelled by WAVERYS and those measured by the buoy network. The differences are always less than 1s (actually a few dozen milliseconds), except in the Bay of Biscay, along the Spanish coasts and at the Azores where the average differences are between 1 and 3s. In the North Pacific and Caribbean Sea, the periods are slightly overestimated, while this is the opposite for the Northwest Atlantic, the Mediterranean and the seas north of Europe (North Sea and Baltic).

Figure 14 and Figure 15 are the RMSD and SI maps computed for the mean wave period with respect to the observations. The two figures are very similar to each other in term of spatial distribution of errors, which underlines that the normalization does not impact too much the SI value, except for the North Sea and the American West Coast. If the scatter index is looked at, lower errors are found again for the tropics, with SIs around 10% for the mid latitudes that score SI from 20 to 30%. As with the SWH, critical locations are in the North Sea for the Azores and within Canada's great lakes.

Figure 16 shows a correlation map. The correlations are slightly lower than what was previously obtained for SWH, but equally satisfactory between 0.6 and 0.9. It can be noted that the lowest correlations are in the extra-tropical belt.

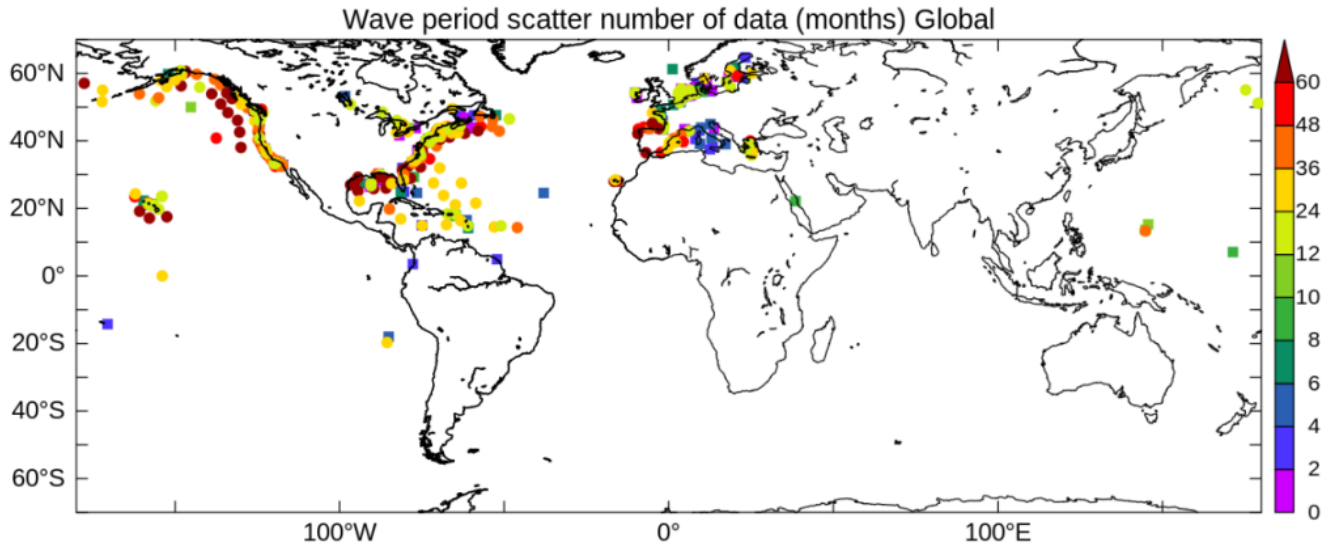


Figure 12. Locations of CMEMS wave buoys, as well as the duration of measurements (colours, in months) for PWP. Time series less than one year are marked by a square instead of a circle.

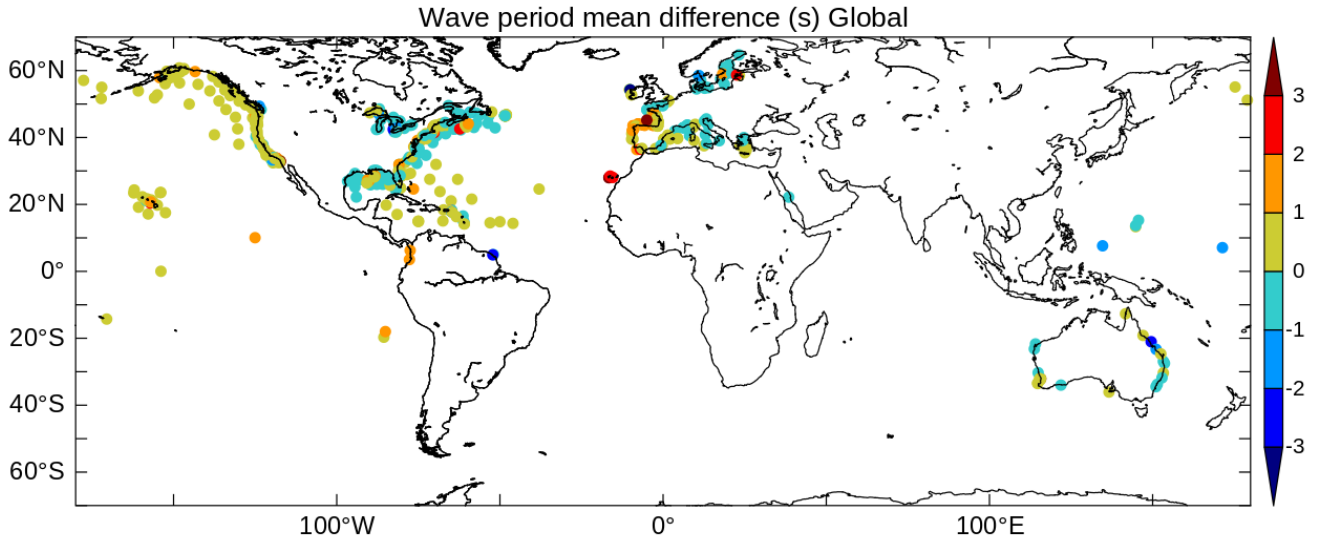


Figure 13. Mean differences in PWP (s) between WAVERYS and CMEMS buoys averaged over the observed period.

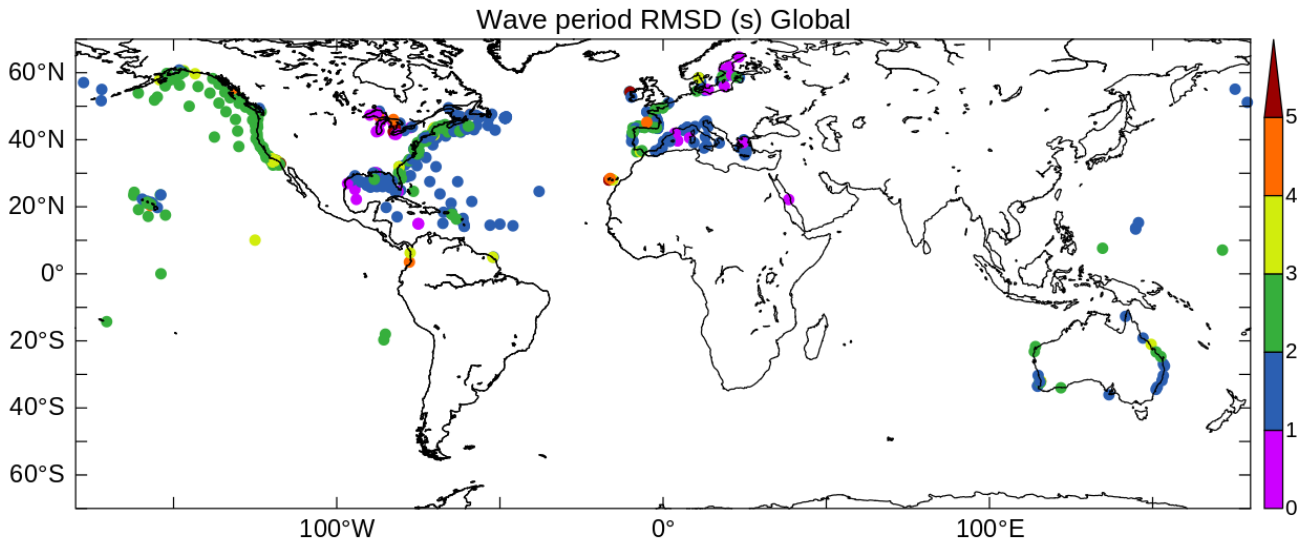


Figure 14. RMSD (m) for WAVERYS PWP with respect to CMEMS wave buoys.

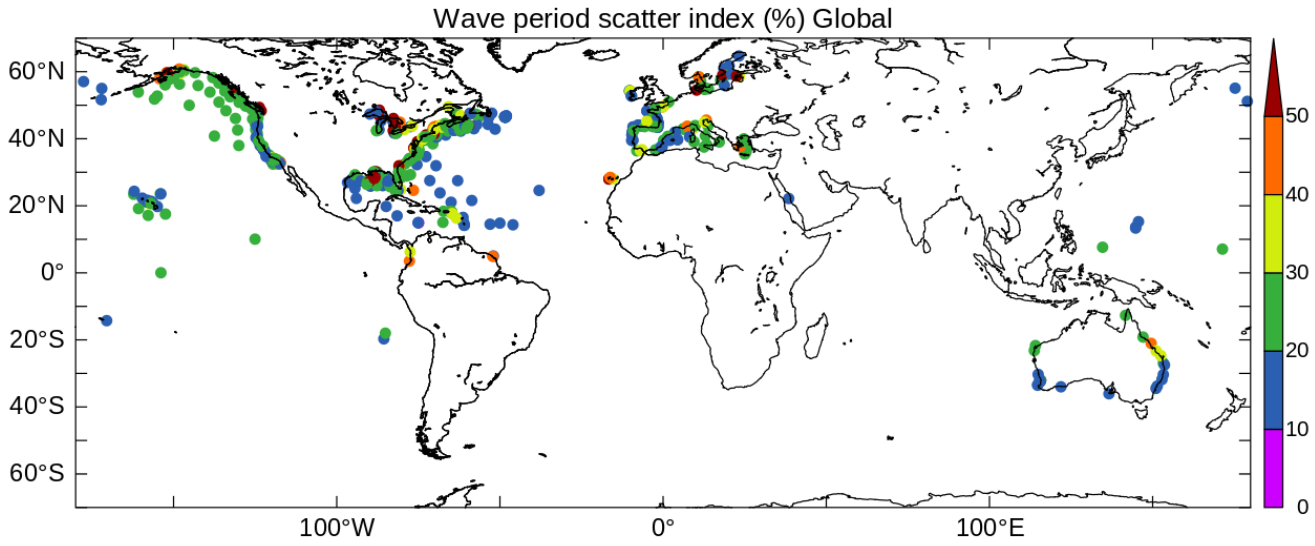


Figure 15. PWP scatter index (%) for WAVERYS against CMEMS wave buoys.

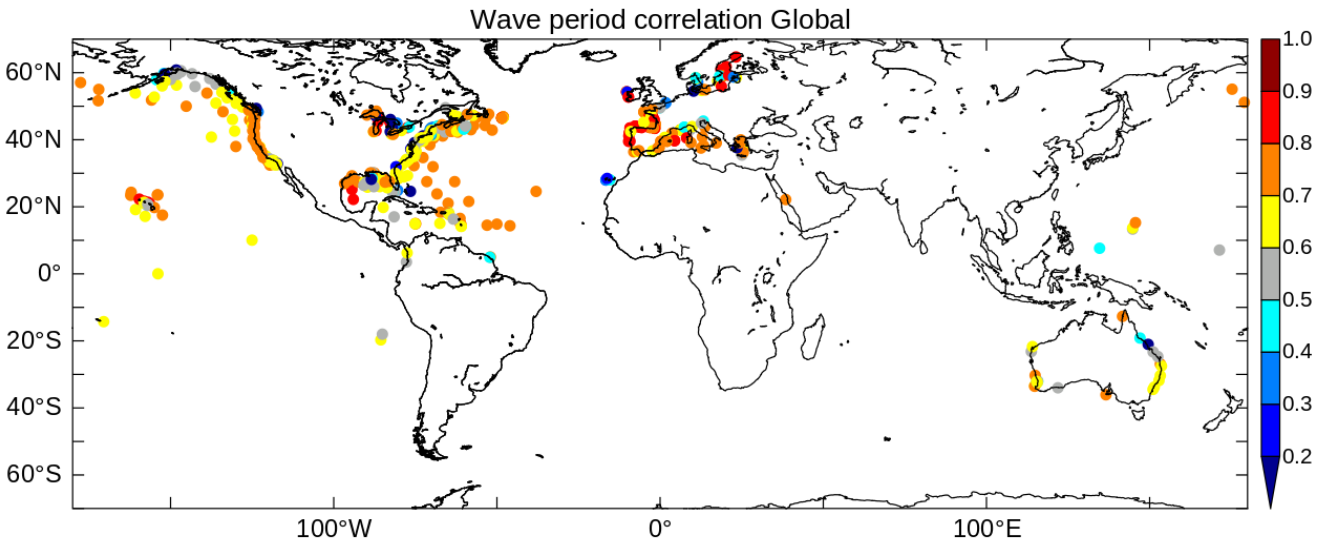


Figure 16. Correlation coefficient of PWP of WAVERYS and CMEMS wave buoys.

presents the results obtained for the mean wave period for different ocean locations. For all buoys a good performance of WAVERYS was noticed with a positive bias of 0.16 seconds and a scatter index of about 26%. Those levels of performance are generally obtained for intermediate latitudes. At the European Atlantic coast, the bias increases to 1 second, but the scatter index and RMSD remain of the same order of magnitude as in other regions. In contrast, along the US east coast the bias is significantly decreased to 0.05 seconds.

shows a different way to analyse the results by using the Taylor diagram. Shelf measurements are also separated from open sea measurements (depth greater than 200 m). Beyond the local physics, the poor results of the buoys anchored in the North Sea and the Baltic Sea can also be explained by their coastal situation, as they stand out very clearly from the others buoys on . Studying only offshore buoys shows that the variability observed for these sites is reproduced quite effectively (the points are located on the standard deviation line of radius 1).

Table 6. Statistical analysis of second order peak wave period for different ocean regions in comparison with in-situ buoys. The Period is set from 1994 until 2015.

Region	Difference (s)	Scatter Index (%)	RMSD (s)	Density of data
Global	0.36	26.2	2.30	11 054 730
High latitudes	0.34	27.2	2.41	1 231 270
Intermediate latitudes	0.36	26.3	2.31	9 030 240
Tropics	0.24	22.7	1.96	793 220
North America West coast	0.49	24.1	2.63	3 263 226
Hawaii	0.42	21.3	2.27	428 228
North America East coast	0.09	28.4	2.17	1 906 719
Gulf of Mexico	-0.18	19.8	1.15	647 195
Caribbean Sea	0.30	20.5	1.67	304 268
Med Sea	0.12	22.2	1.20	823 289
Europe Atlantic coast	1.13	24.9	2.40	1 209 937
North Sea	0.34	34.1	2.17	4 867
Baltic Sea	-0.31	34.3	1.70	196 416

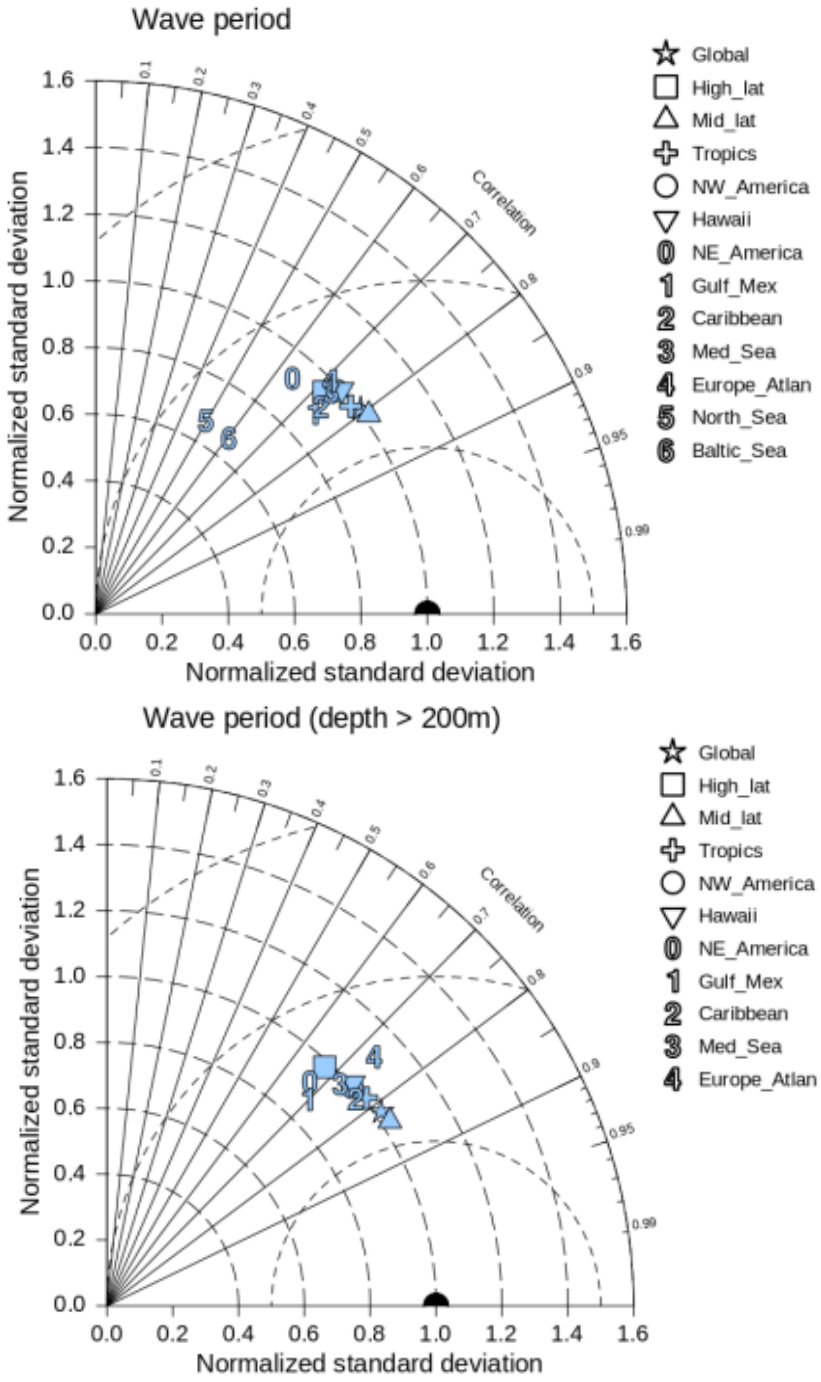


Figure 17. Normalized Taylor diagrams of PWP compared with buoys. Statistics for different regions at all depths (top) and depths deeper than 200 m (middle) or shallower than 200 m (bottom) (continues in next page).

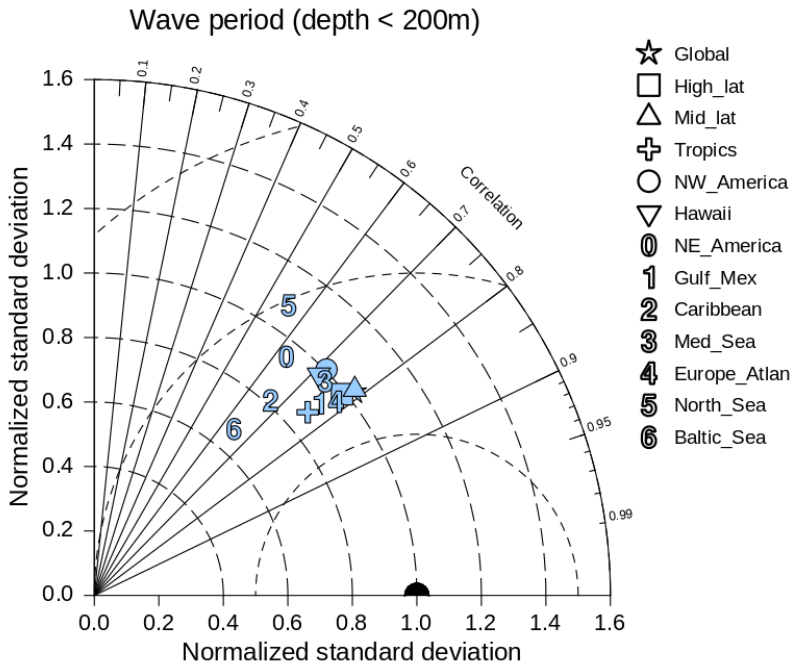


Figure 17. (cont.) Normalized Taylor diagrams of PWP compared with buoys. Statistics for different regions at all depths (top) and depths deeper than 200 m (middle) or shallower than 200 m (bottom).

IV.1.5 Inter-annual consistency: comparison to ERA5 wave climatology

As previously approached, the mean peak period biases were compared against the ERA5 wave reanalysis (Figure 18). This figure shows similar behaviours as those observed when comparing significant wave heights (i.e., global biases of about 0.25 s until 2005), then almost zero differences for the rest of the series. The scatter index is constant and around 22.5%.

Figure 19 shows that the differences with ERA5 are mainly localized in the equatorial Pacific due to the propagation of long swells of subpolar origin (likely because of differences in the WAVERYS and ERA5 dissipation source terms). A negative bias of a few seconds can also be noted compared to ERA5 located in the Agulhas current area and in the North Indian. ERA5 does not have current forcing and the other results suggest that WAVERYS is then more efficient in this region.

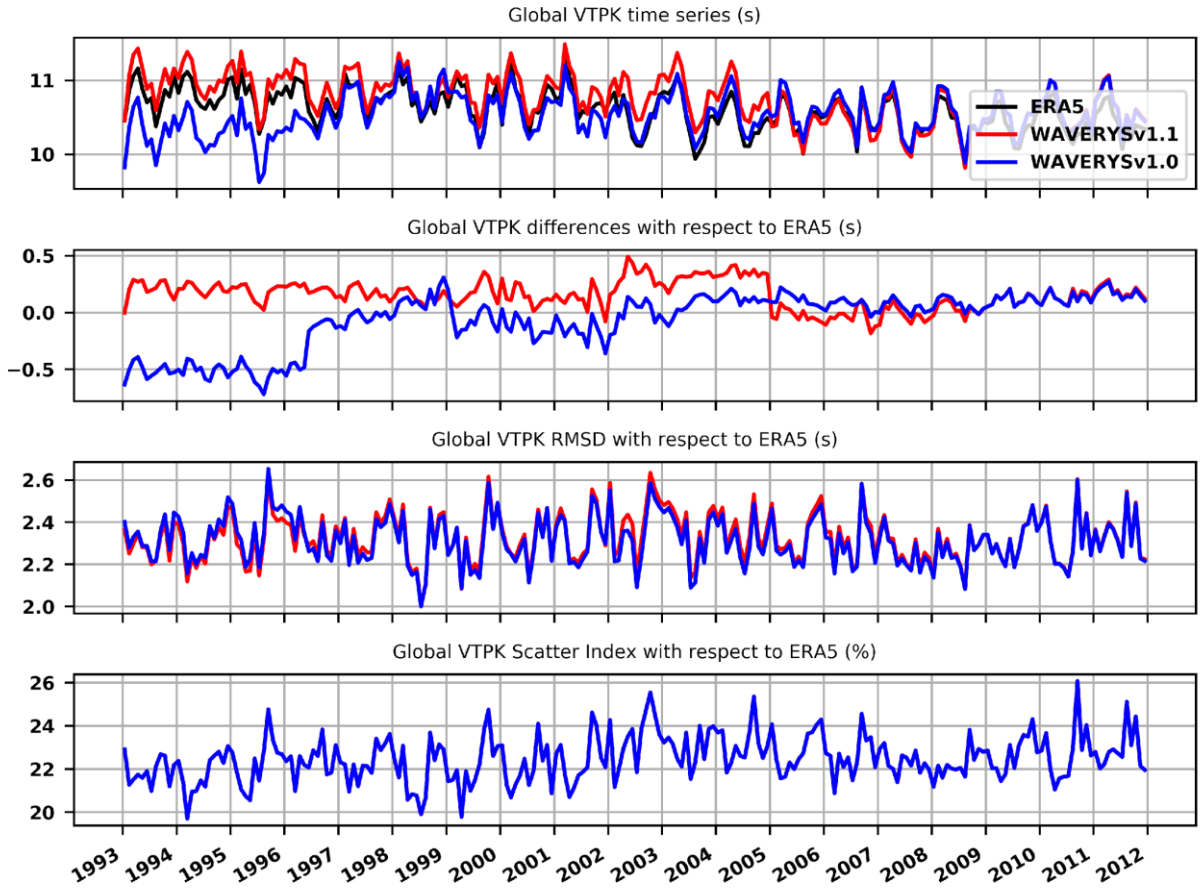


Figure 18. Comparison of the PWP (Peak wave period) time series of WAVERYS and ERA5.

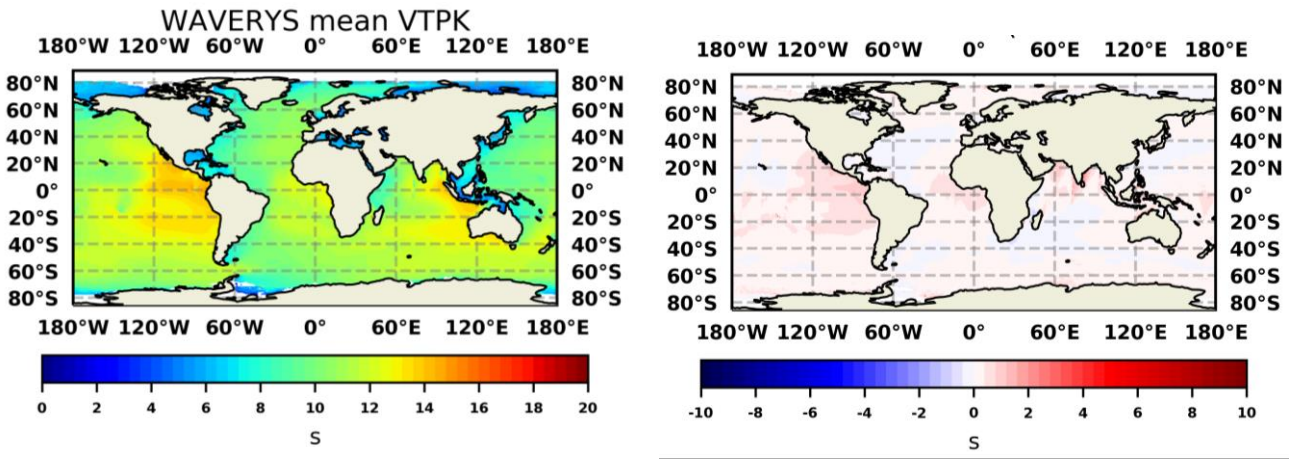


Figure 19. Left panel: mean PWP (s) from WAVERYS for the whole period. Right panel: average difference in peak wave period between WAVERYS and ERA5 for the whole period.

IV.3 Mean wave period

IV.1.6 Coastal and in-situ validation: comparison to CMEMS in situ data

In this section the second order mean wave period of WAVERYS and CMEMS in-situ wave buoys are compared. It can easily be seen that WAVERYS slightly underestimates the mean wave period except for Galician coast and the Bay of Biscay where the wave period is slightly overestimated (Figure 20).

The results are also satisfactory in terms of RMSD (Figure 21), with values less than one second at most sites except on the west coast of the United States and in the middle of the Pacific Ocean (Northern Mariana Islands). This shows that the variability of the average periods, which may be related to the arrival of different swells or waves raised by the local wind is well perceived by the model.

RMSD is satisfactory with values of less than 1 second at most buoys except on the US west coast and in northern Marian islands in the Pacific Ocean. This shows the variability of the average periods, which may be related to uncertainties on the wind forcing for the swell propagation at the beginning from storms in the Southern Ocean or waves generated by the local wind.

Table 6 presents the results obtained for the second order mean wave period for different ocean locations. For all buoys we notice a good performance of WAVERYS with a scatter index of about 15%. However, the mean wave period is underestimated by WAVERYS with a bias of -0.4 seconds. The best performance of WAVERYS is obtained for the tropical and Hawaiian buoys with a scatter index of approximately 11 %. These two ocean areas are swell dominant sea-state and this confirms that WAVERYS has a good accuracy for capturing swell in the Pacific Ocean. One can notice that the bias is decreased at the European west coast with a slight overestimation of the mean wave period of WAVERYS. We also observed more scatter for the mean wave period at the North Sea area,

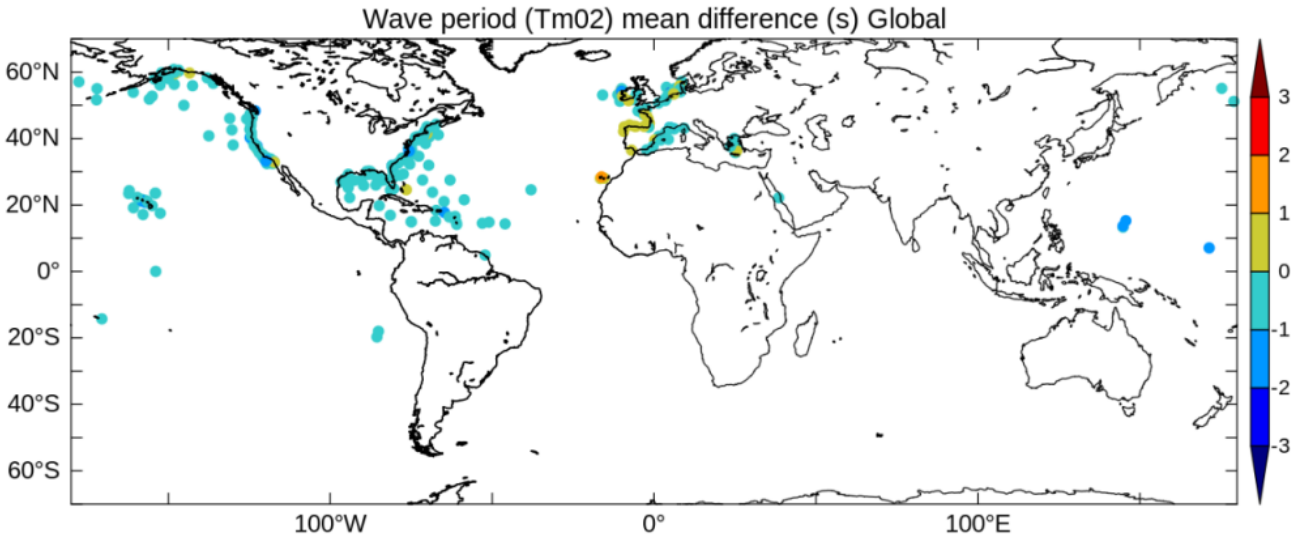


Figure 20. Bias of mean wave period (s) of WAVERYS with respect to CMEMS buoys averaged over the observed period.

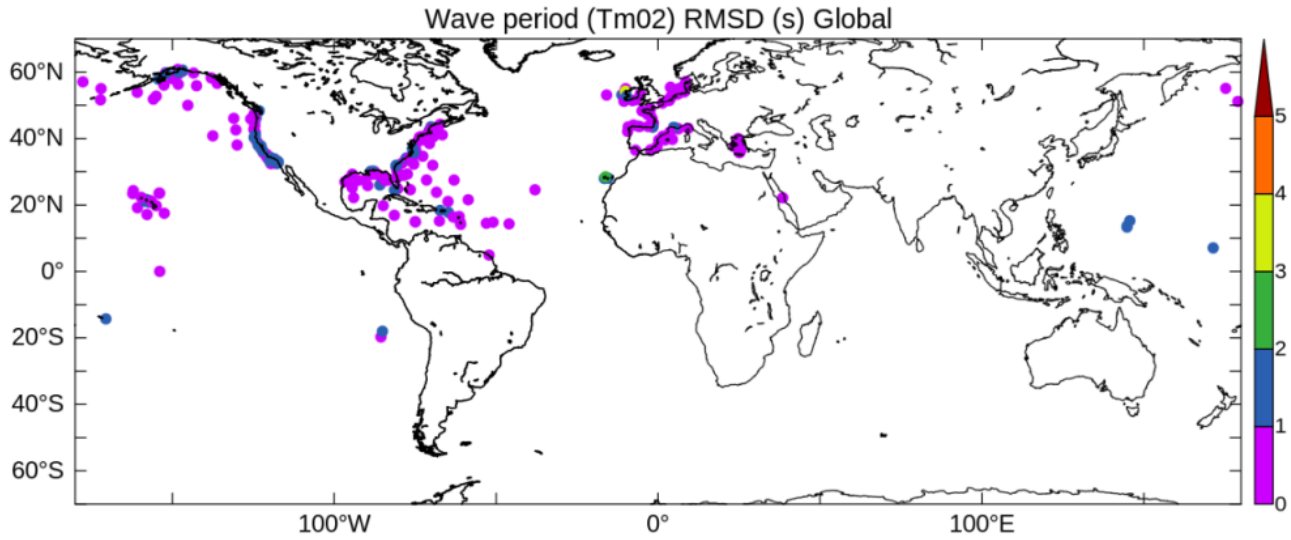


Figure 21. RMSD of mean wave period of WAVERYS in comparison with CMEMS wave buoys.

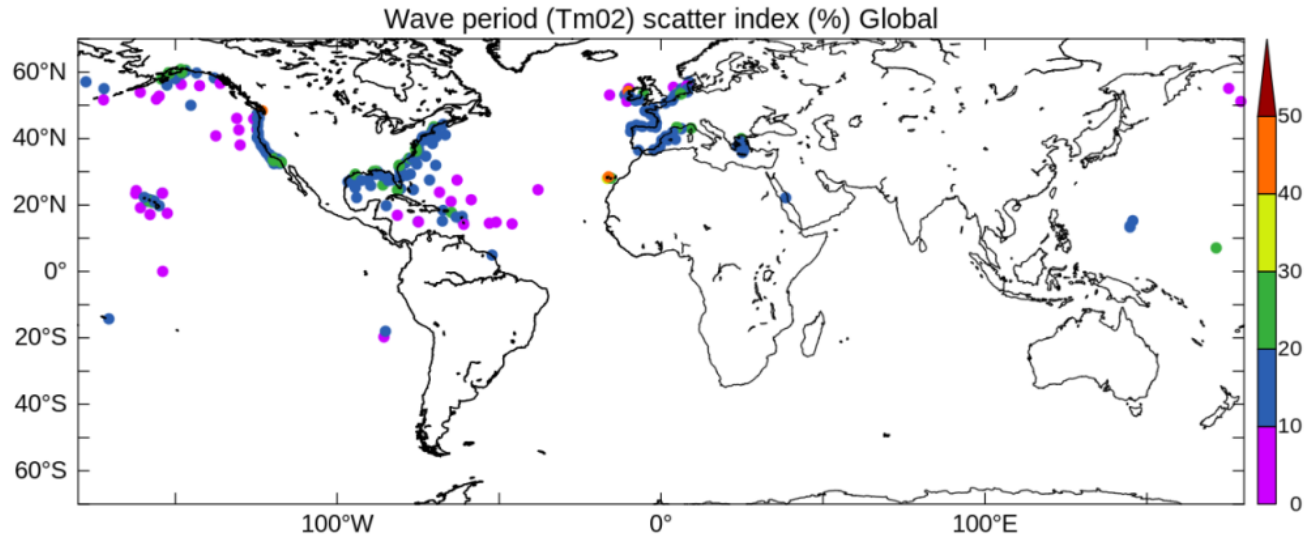


Figure 22. Scatter index of mean wave period (%) of WAVERYS in comparison with CMEMS wave buoys

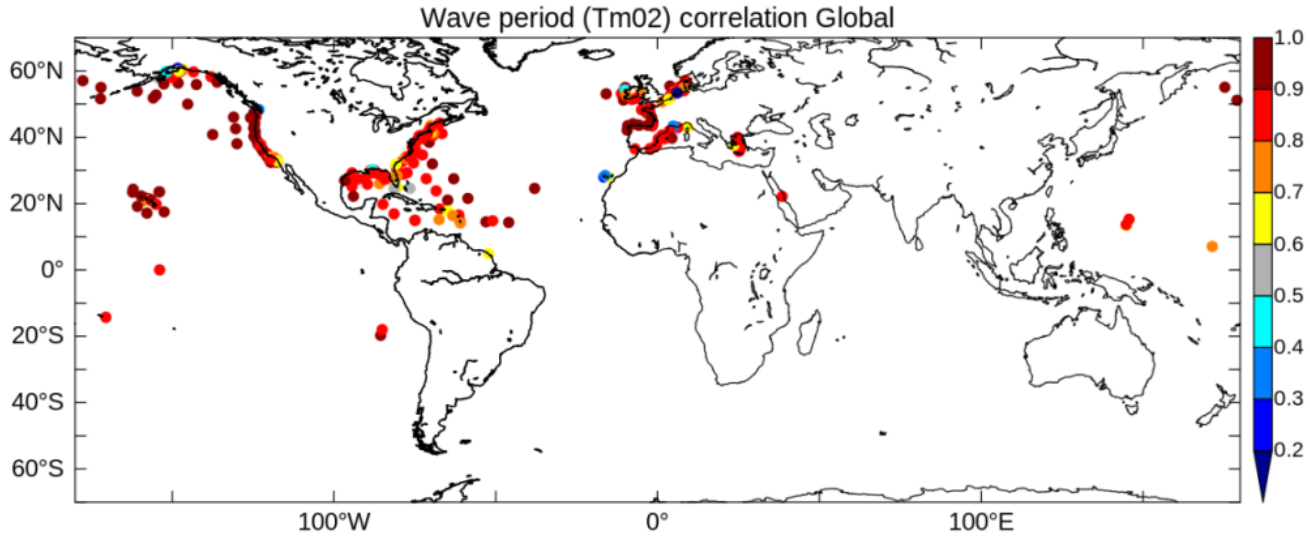


Figure 23. Correlation between mean wave period time series from WAVERYS and CMEMS wave buoys.

Table 7. Statistical analysis of second order mean wave period (s) for different regions by comparison with in-situ buoys. Period: 1994-2015.

Region	Difference (s)	Scatter index (%)	RMSD (s)	Density of data
Global	-0.31	14.8	0.86	8 745 364
High latitudes	-0.29	12.4	0.76	940 331
Intermediate latitudes	-0.31	15.4	0.88	7 181 273
Tropics	-0.38	11.4	0.69	623 760
North America West coast	-0.41	13.1	0.91	2 631 756
Hawaii	-0.40	12.0	0.81	4 086 694
NE America	-0.55	16.4	0.86	1 515 901
Gulf of Mexico	-0.65	17.7	0.79	699 025
Caribbean Sea	-0.30	10.6	0.59	260 096
Med Sea	-0.24	14.4	0.57	798 718
Europe Atlantic coast	0.17	14.7	0.89	13 543 661
North Sea	-0.14	16.7	0.69	4 867

IV.4 Mean wave direction (MWD)

Wave direction follows the classic meteorological convention: 0° corresponds to north, 90° to east, etc., with the direction indicating the direction of origin. **NB: There is an offset of 180° in the netcdf format attributes** (normally taken into account automatically by netcdf compatible software).

IV.1.7 Coastal and in-situ validation: comparison to CMEMS in situ data

The number of buoys measuring wave direction is even smaller than the data set providing the wave period (Figure 24). The European network remains almost unchanged with respect to wave period and the North American network loses many measurement points.

Figure 25 shows the average directional deviations between WAVERYS and the available buoys. Following the convention of CMEMS wave files (0°=North, 90°=East), a positive (respectively negative) difference means waves incoming too far from the left of the buoy (respectively right). Overall, propagation deviations are small on average, in the order of -5°/+5°. Regional biases can be noted, as in Hawaii (+15° difference), the West American coast (+5° difference), the Eastern Mediterranean (+15° difference) and the Gulf of Alaska (-10° difference).

Figure 26 shows the RMSD on wave train directions. The areas with a satisfactory RMSD (less than 30°) are in the Bay of Biscay, off the coast of Ireland and in the Caribbean. The less precise areas in terms of variability are again the Cape Verde and Azores islands (direction RMSD around 80°).

Figure 27 shows the correlation of WAVERYS in terms of direction with respect to the buoys. This diagnosis aims to verify if the arrival of wave trains from different sources at the measurement point are well phased with observation (a simple delay of a well modelled wave train is then perceived as a defect). This time, high and mid-latitudes are better than the tropics in terms of direction (correlation above 0.6 vs. correlation below 0.5). This is likely due to the propagation of swells well captured by the wave

model. Very low correlations (> 0.2) are found in the tropical Atlantic, French Polynesia, Hawaii, and off California.

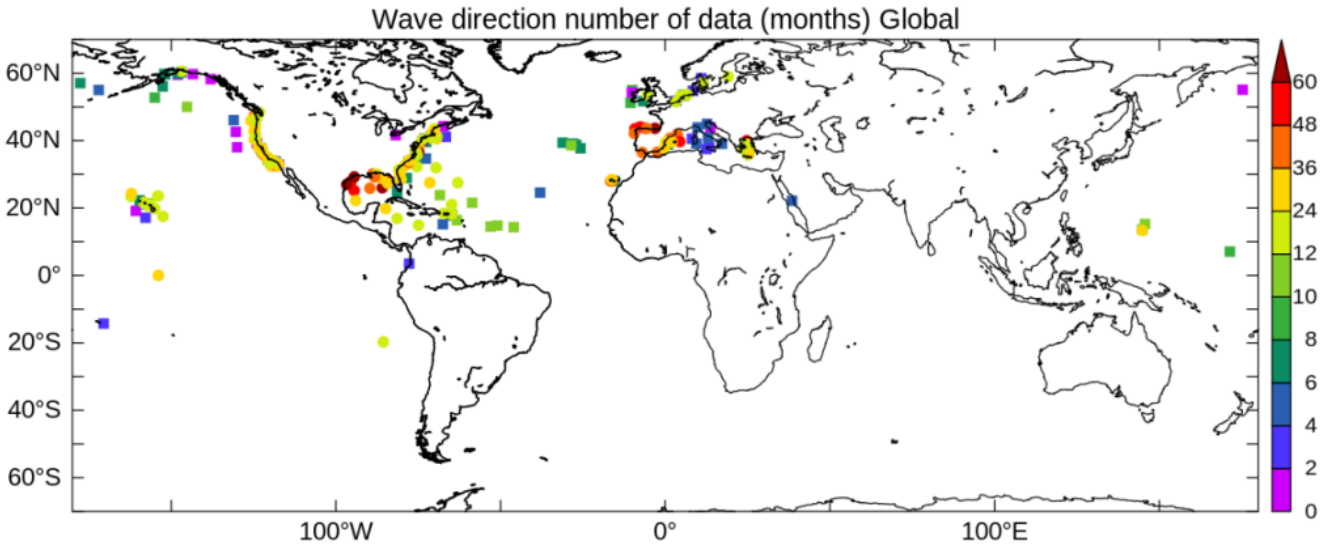


Figure 24. CMEMS wave buoy locations and the duration of recorded measurements (colours, in months) for MWD. Time series less than a seasonal cycle (one year) are marked by a square instead of a circle.

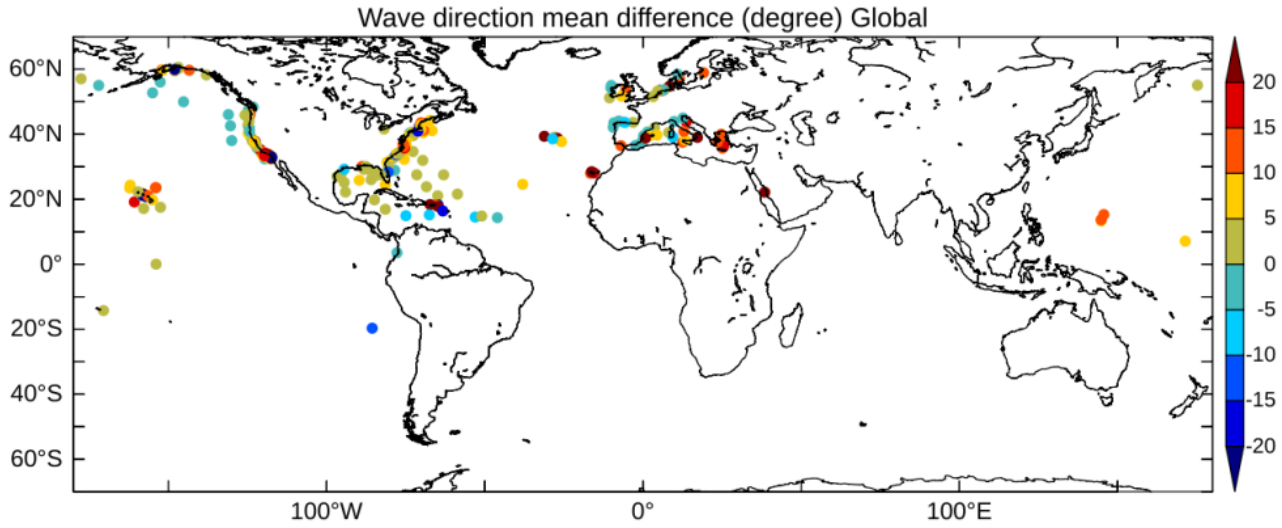


Figure 25. Mean difference MWD in degrees between WAVERYS and CMEMS buoys over the observed period.

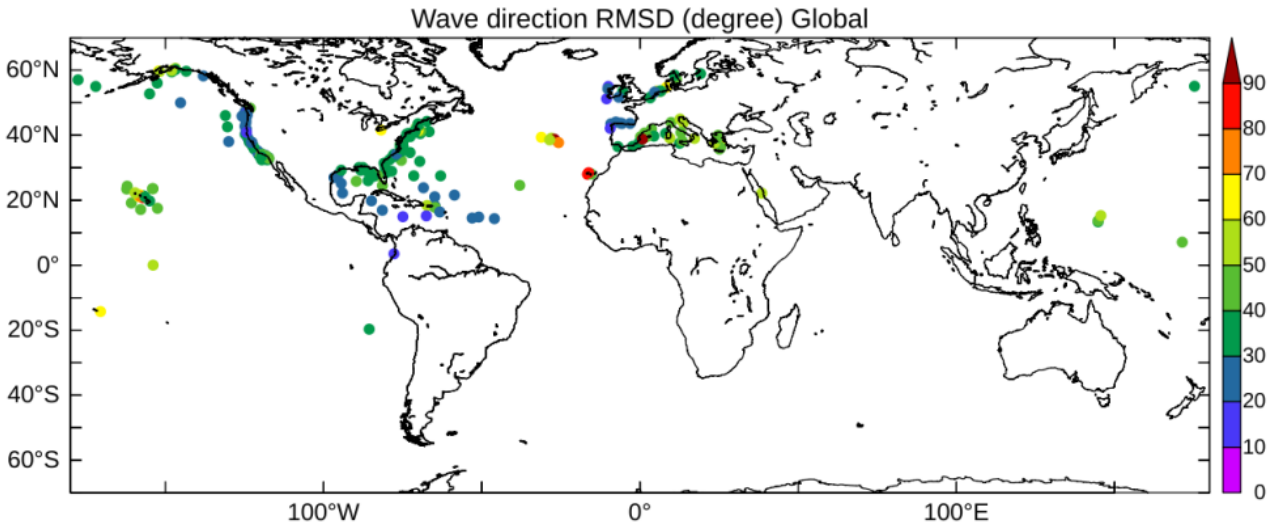


Figure 26. RMSD of MWD in degrees for WAVERYS with respect to CMEMS wave buoys.

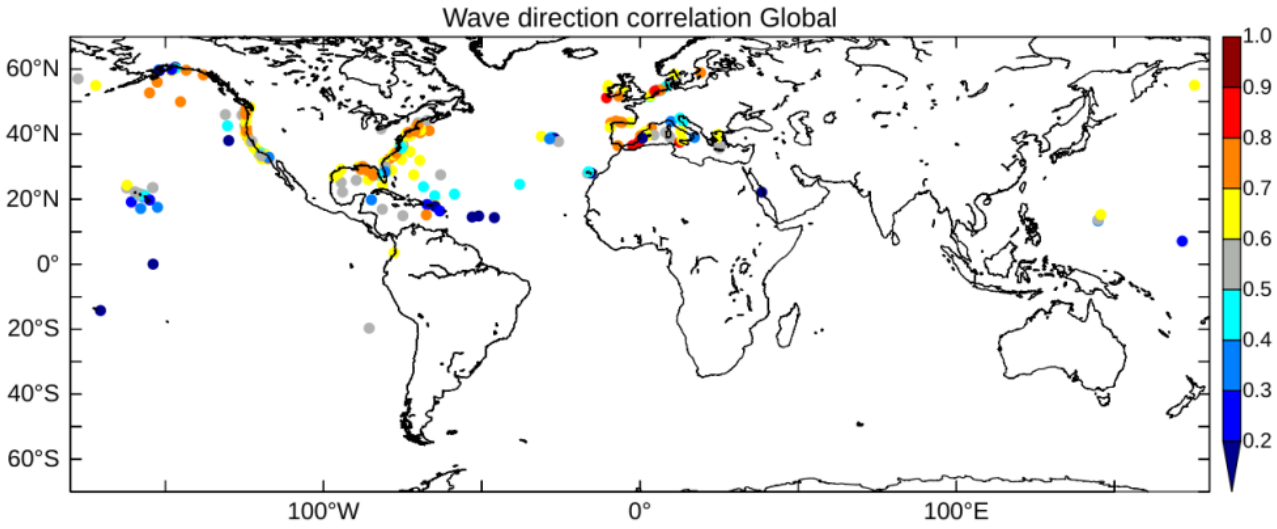


Figure 27. Correlation coefficient of MWD between WAVERYS and CMEMS wave buoys.

shows the results of the comparison between MWD from WAVERYS and CMEMS wave buoys. Considering all buoys, the bias and RMSD are satisfactory and estimated to 5° and 38° respectively. The same tendency was noticed when looking especially at high and intermediate latitudes. The average difference of MWD between WAVERYS and buoys are less than 10°, except for the Baltic Sea (10.9°) and the Azores (13.61°). The RMSD of MWD for the different ocean areas is in general between 25 and 40°, except for the Azores (70°), Hawaii (55°), the Mediterranean Sea (47°), and the North Sea (45°).

Figure 28 illustrates the results plotted in a Taylor diagram. It seems that the standard deviation of MWD is better reproduced for buoys located at depth less than 200 m. This can be explained by the straightening of the wave trains parallel to the bathymetric lines in shallow water (depth refraction). On the other hand, the correlation coefficient deteriorated for measurement points in the Azores and the Caribbean.

Table 8. Statistical analysis of direction for different regions by comparison with in-situ buoys (Period 1994-2015).

Region	Difference (°)	RMSD (°)	Density of data
Global	4.98	37.8	3 340 640
High latitudes	3.9	42.37	213 099
Intermediate	5.3	37.49	2 887 385
Tropics	1.98	37.15	240 156
North America	6.11	30.53	985 801
Hawaii	3.32	55.09	152 052
North America	5.51	37.32	577 438
Gulf of Mexico	2.64	33.01	422 762
Caribbean Sea	-0.99	31.02	104 868
Azores	13.61	70.31	26 990
Med Sea	7.94	47.06	363 818
Europe Atlantic	1.07	25.07	295 098
North Sea	3.61	45.03	92 792
Baltic Sea	10.98	36.5	28 269

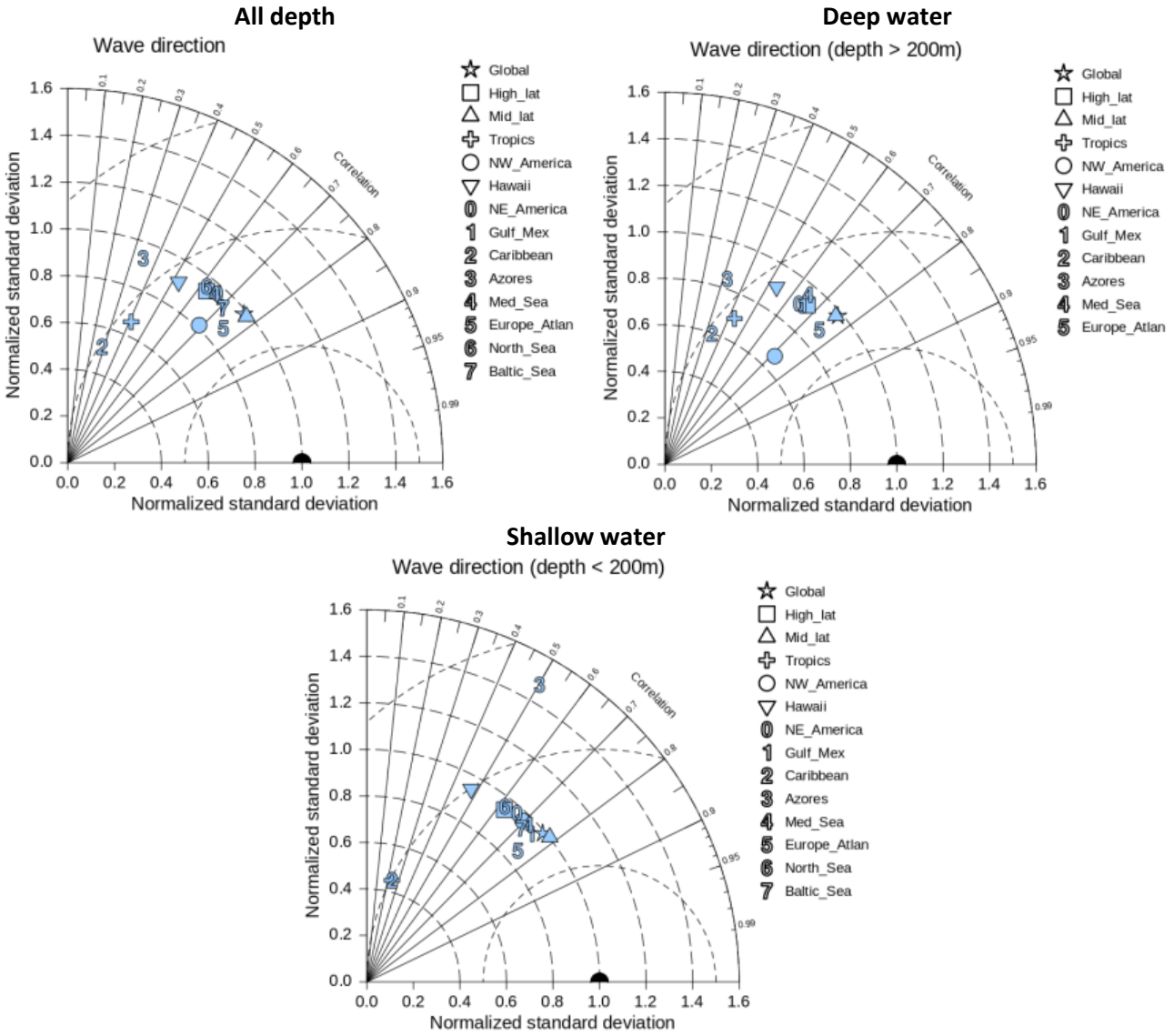


Figure 28. Normalized Taylor diagrams of MWD of WAVERYS in comparison with buoys. Statistics for different regions at all depths (top left) and depths > 200 m (top right) or < 200 m (bottom).

IV.1.8 Inter-annual consistency: comparison to ERA5 wave climatology

The top of Figure 29 shows the average wave directions calculated from WAVERYS for the whole period. The wind-waves trains linked to the westerly winds and the Trade Winds can clearly be seen. The lower figure shows the average wave direction deviations calculated every 3 hours between WAVERYS and ERA5, then averaged over 1993-2018. The average deviations of average directions are less than 20° (the angular resolution of both models is 15°). Differences in current systems can also be reported, particularly for the Kuroshio Current and the North Atlantic drift.

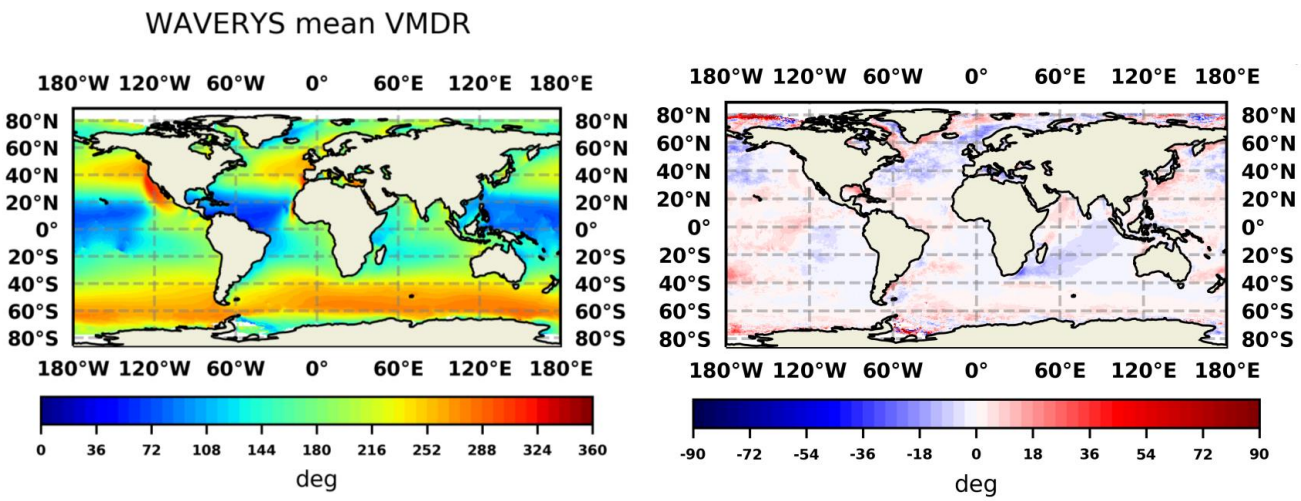


Figure 29. Left panel: Average of mean wave direction (degrees) from WAVERYS for the whole period. Right panel: Average difference of mean wave direction (degrees) between WAVERYS and ERA5 for the whole period.

IV.2 Wave rose diagram

The wave rose diagram is an interesting validation tool which can provide the dominant wave regime depending on the energy, direction, and frequency of waves present at the selected ocean location. In this section the wave roses calculated from CMEMS in-situ data and the ones obtained from WAVERYS were compared. Figure 30 illustrates for instance the results for selected buoys. Overall, the most prominent behaviours of current roses are reproduced with WAVERYS.

The first example is a mooring off the north-western tip of California. The regular wave regime is a swell centred on the southeast and spread over about 90°. WAVERYS fits nicely this wave pattern, even if the spread is more restricted over the Southeast, with more occurrence for this direction. Besides, waves from east and west are not present in the WAVERYS wave rose. There is also consistency between the SWH distributions according to occurrences. The presence of high waves (SWH between 5 to 10 meters) is noted, but of low occurrence. WAVERYS does not resolve those high waves in the east-south-east direction.

The second and third examples, respectively of moorings in the Pacific (Hawaii) and the North Sea, yield similar conclusions. The Hawaiian mooring observes waves from the north and northeast and WAVERYS seems to provide higher number of waves from this direction too. WAVERYS shows less frequent occurrences of easterly waves and more frequent ones from the west, west-southwest, and southwestern waves that are absent in the observations. The third example in the North Sea shows a more complex sea state, with large directional spreading related to the wind sea (presence of short waves < 20 cm) superimposed on a southwest swell. The pattern of wave directions is well represented

by WAVERYS. The magnitude and frequency of short waves (> 20 cm) are overestimated from certain directions.

The fourth example is in the Azores. It has been seen in the previous sections that the wave moorings in this area offered the worst results for WAVERYS. The observed regime is a primary southwest swell that WAVERYS predicts with a too large spread and a secondary west swell that WAVERYS misses.

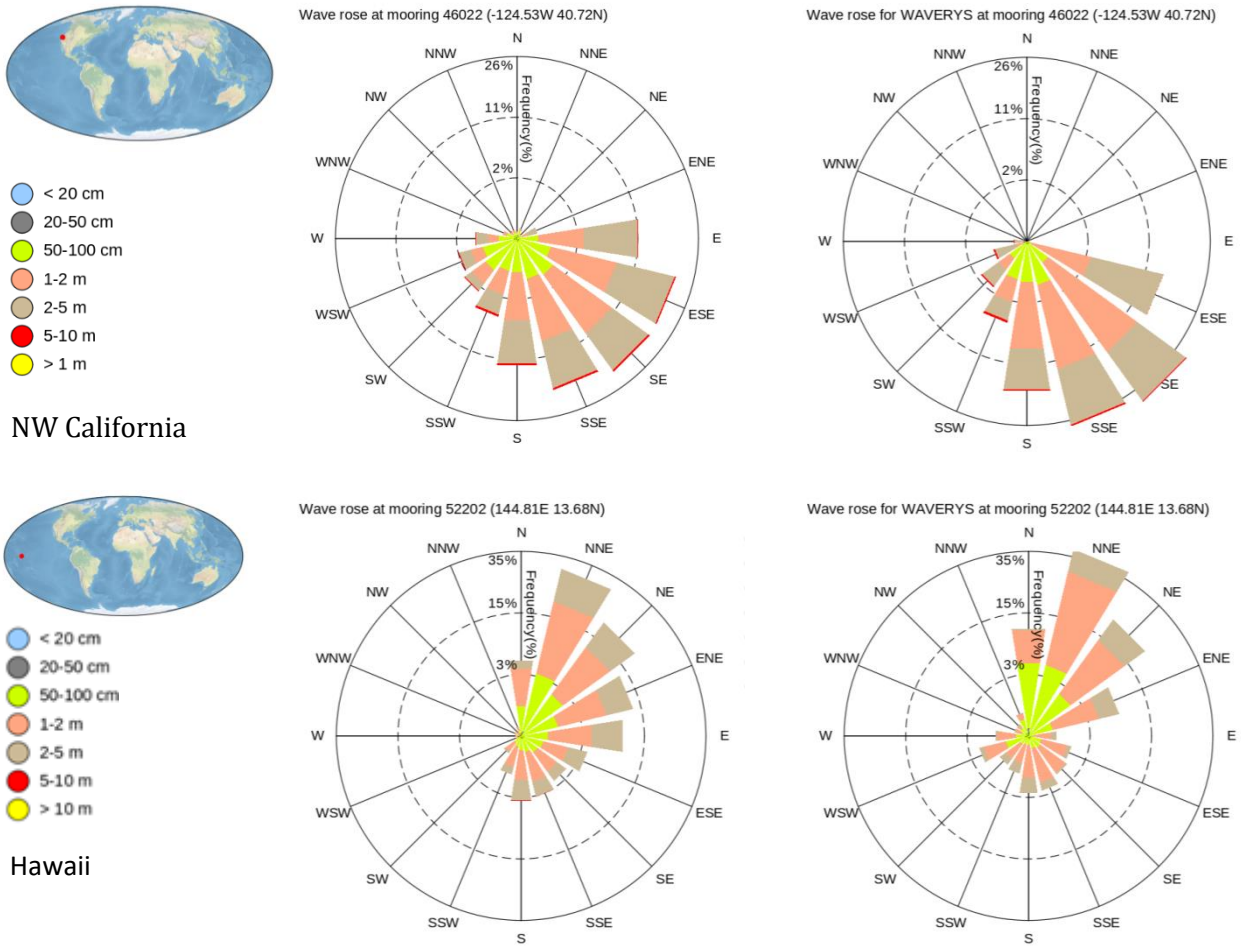
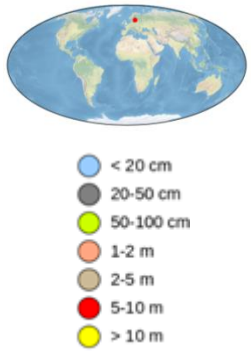
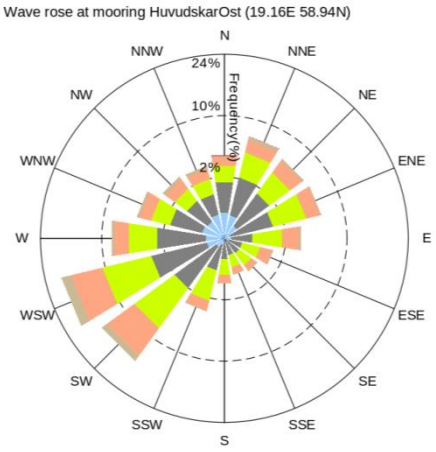
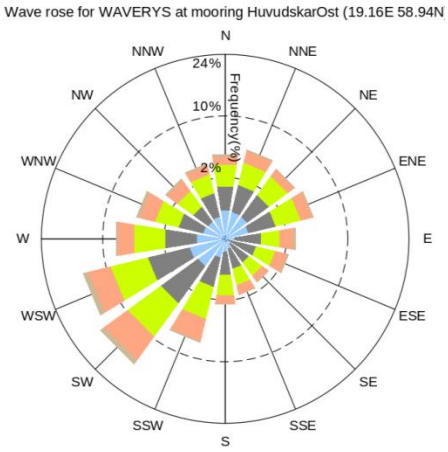


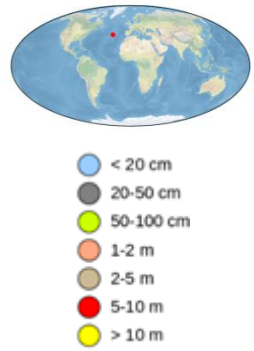
Figure 30. Wave roses computed from the comparison of CMEMS wave buoys and WAVERYS. The map on the left shows the buoy location, while the middle and the left panels stand for buoys and WAVERYS, respectively. The radial scale gives the frequencies of occurrence of wave height/direction couple and is not linear (continues in next page). Edit: the last range of heights for the legend (yellow colour) is indeed greater than 10 m.



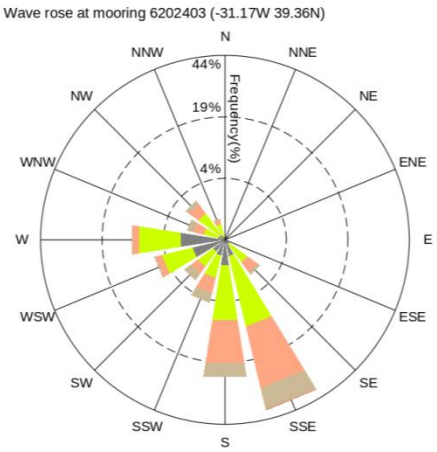
North Sea







Azores



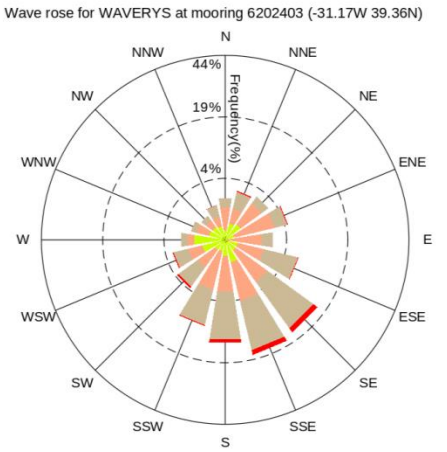


Figure 3030 (cont.). Wave roses computed from the comparison of CMEMS wave buoys and WAVERYS. The map on the left shows the buoy location, while the middle and the left panels stand for buoys and WAVERYS respectively. The radial scale gives the frequencies of occurrence of waves for a particular heigh and direction. Please note that scale is not linear..

V SYSTEM'S NOTICEABLE EVENTS, OUTAGES OR CHANGES

Date	Change/Event description	System version	other
07/2020	Correction of WAVERYS initial time series + 2019 time extension	V1.1	
07/2022	2021time extension + change of model version (ECMWAM cy42 vs cy38 before) + assimilation of SWIM/CFOSAT wave spectra	V1.2	
07/2023	2022 time extension, bug fix on the 2020-2021 period, Interim production since April 2023	V1.2	
07/2024	Extension in time from the 1980's	V1.2	

VI QUALITY CHANGES SINCE PREVIOUS VERSION

VI.1 Version 1.1: 2019 time extension + bug fix (EIS 2020)

VI.1.1 Correction of biases caused by the first altimetry missions

The time series of WAVERYS V1.0 is biased from 1993 to 1996 (about -15 cm on the global SWH for example). This is due to the use of uncorrected altimetry data over this period and this bias can be clearly seen in comparisons to time series of ERA5 waves in Figure 32.

The significant height assimilated by the first altimeter missions tends to be negatively biased, which introduces some energy deficit compared to the rest of the period and to the ERA5 time series over the first years. The problem is most visible for the years 1993-1997, but satellite data have been reprocessed until 2005 to ensure a smooth transition over the time series. Figure 32 shows the effect of this correction on global time series derived from the significant wave height for V1.0, V1.1, and ERA5. Figure 32 also shows that this correction has practically no effect with validation quantities that deals with variability, like RMSD or scatter-index (with respect to ERA5).

VI.1.2 Correction of assimilation for some altimetry missions

A bug in the assimilation system for V1.0 resulted in spurious and fuzzy analysis increments of significant wave height along-track of some specific satellite missions. The missions concerned are as follows: JASON1, JASON2, and CRYOSAT. The bug is not systematic but may appear from time to time during the years 2002-2016. The differences are oscillations of Hs of the order of about 10 cm underneath satellite tracks.

VI.1.3 Correction for first swell period

An error had crept into the model output post-processing program that caused the minus one-moment period to be broadcast in V1.0 instead of the first moment period for the primary wave period. The issue is now corrected in V1.1.

VI.1.4 Impact of the V1.1 upgrade on validation scores

The validation carried out with the inshore buoys for V1.1 series showed very little changes from previous results. This is not surprising, as the buoys are mostly located close to the coast, far from the traces of satellite assimilation which takes place in the open ocean. The Taylor diagram for Hs errors in Figure 32 compared to the one presented in Figure 10 for version 1.0 has not significant differences. In average, computing the global scatter index and regression slope show also very little differences with what previously obtained on

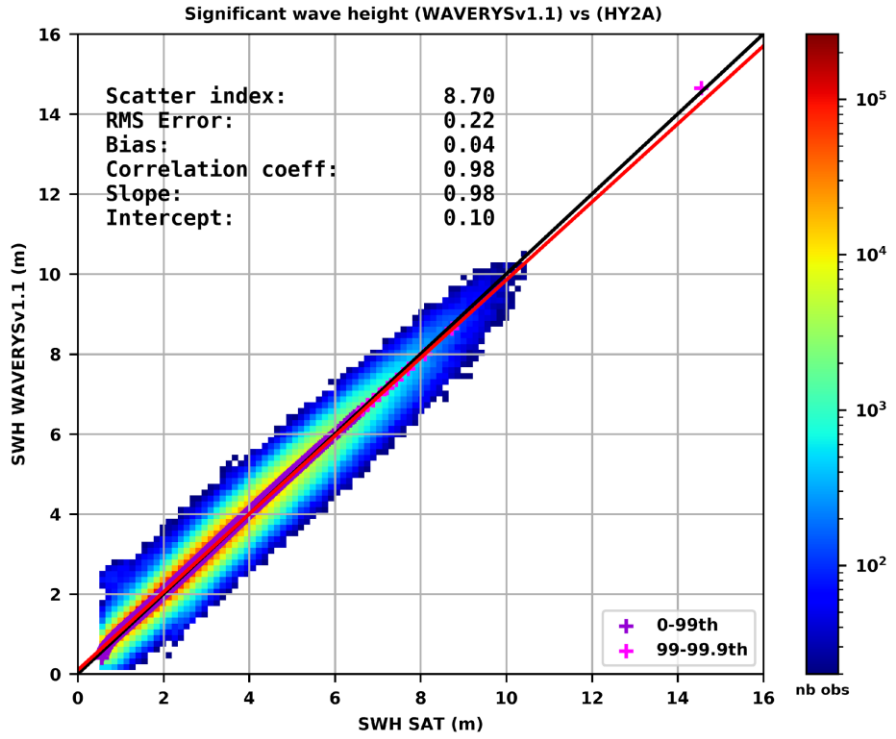


Figure (V1.1 [SI of 8.7%, slope of 0.98, bias of 0.4 m] vs V1.0 [SI of 8.83%, slope of 0.97, bias of 0.3 m). Differences should be expected in the very vicinity and at the time of inconsistent data assimilation as shown in Figure 31.

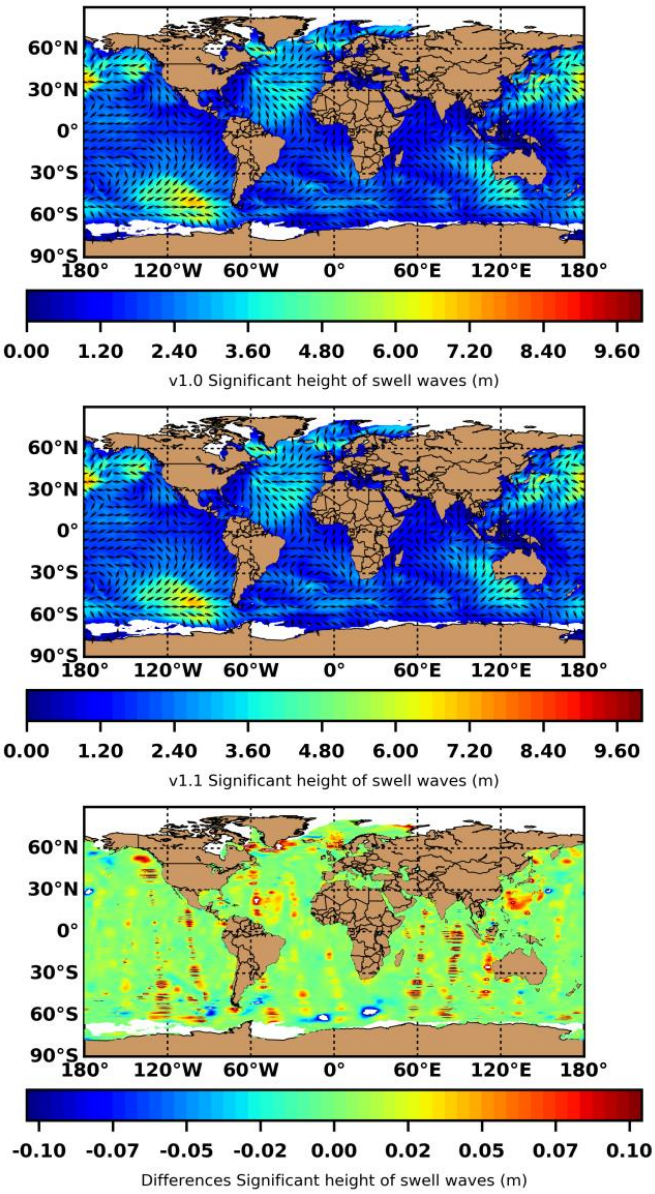


Figure 31. top: SWH in v1.0 for a special date, middle: SWH for the same date for v1.1; and bottom: differences between the two versions that exhibits satellite tracks.

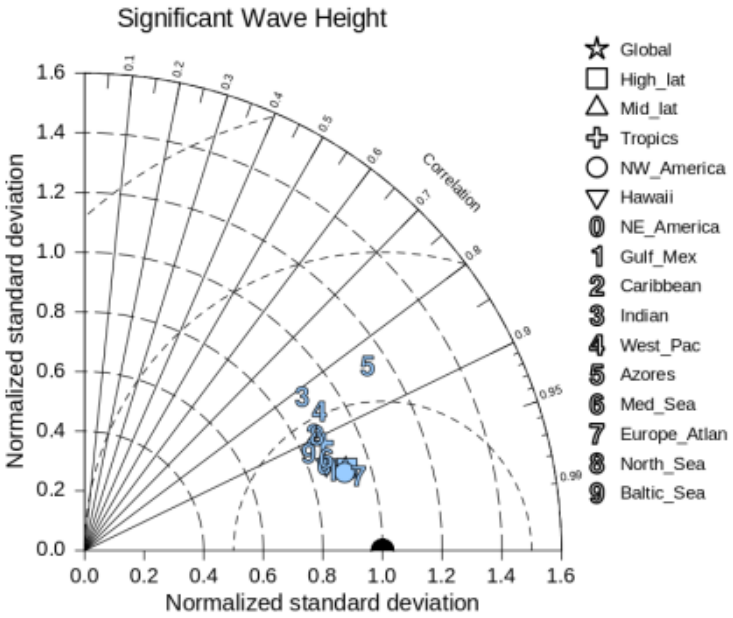


Figure 32. Up: Coastal buoys validation for WAVERYS V1. :

VI.2 Version 1.2: 2020-time extension (EIS 2021)

Version 1.2 encompasses the 2020 time extension of WAVERYS. The version number change is mostly linked to the model upgrade (cy42 version from IFS against cy38 version previously). The change version for the model's core does not represent a physical improvement as such, as it just consists in numerical and computational improvements. We can nevertheless note the assimilation of SWIM spectra from February 2020 as a physical upgrade from the previous WAVERYS time series. Figure 33 shows the validation of the temporal extension for the months of January and February 2020 from HY2-b data (independent altimeter for WAVERYS). We notice a slight underestimation of heights for winter storms in the northern hemisphere. For wave heights below 8 m, we note a very good agreement between WAVERYS and HY2-B observations. The 2020 time extension should yield error ranges equivalent to the most recent years of WAVERYS.

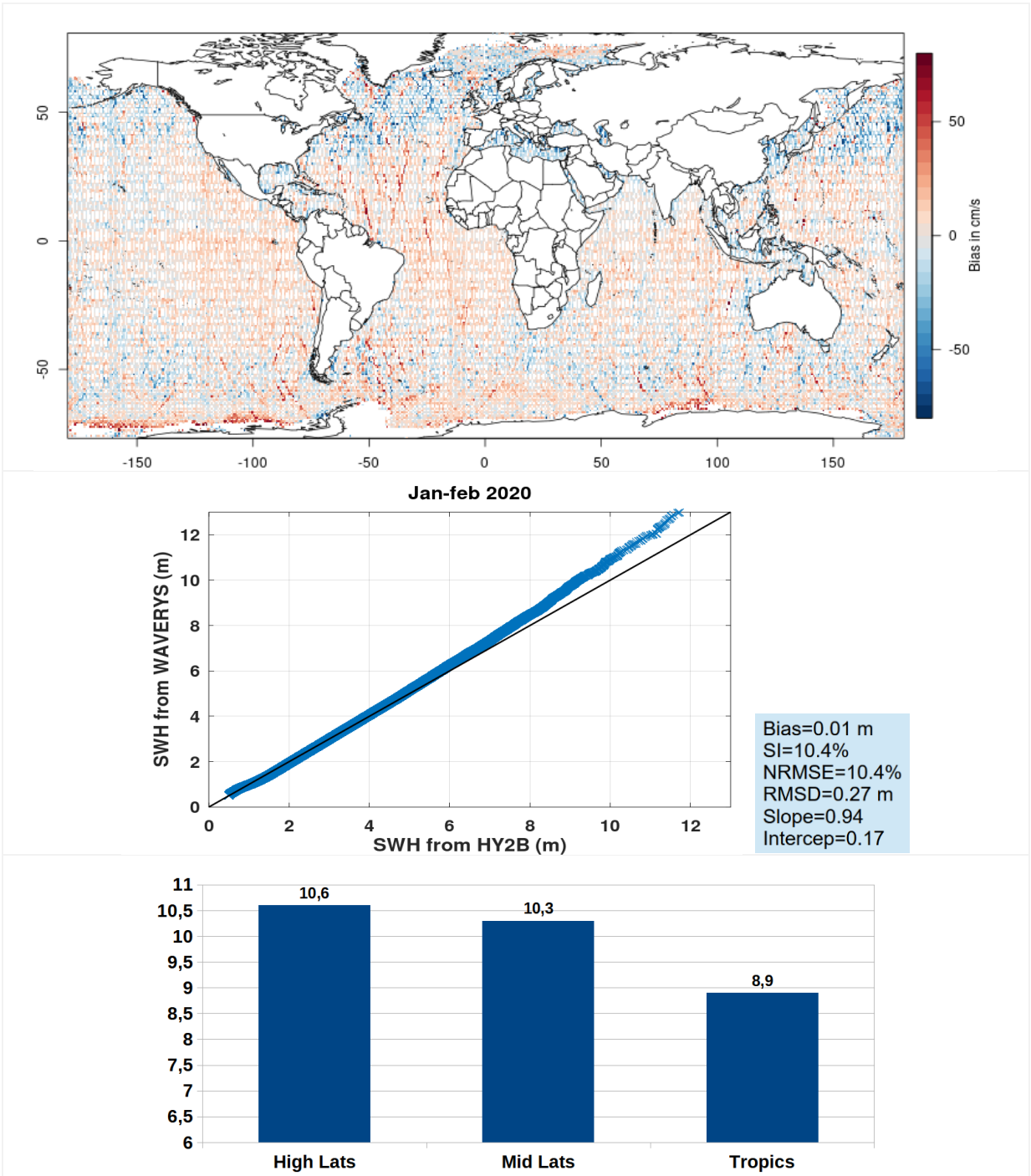


Figure 33. Top: Average SWH bias map for the months of January and February 2020. The reference observations are those of the HY2-b altimeter which is not used in WAVERYS' data assimilating system. Middle: Scatterplot diagram between SWH modelled from WAVERYS and measured by HY2B for Jan-Feb 2020. Usual statistics can be found at the right of the plot. Bottom: Average scatter index value per region for Jan-Feb 2020.

VI.3 2022 time extension + interim production + bug fixes (EIS 2023)

VI.3.1 2022-time extension + real-time catch-up for Interim production

EIS 2023 offers a real-time catch-up of the original WAVERYS dataset "***cmems_mod_glo_wav_my_0.2deg_PT3H-i***" to the start of the interim dataset "***cmems_mod_glo_wav_myint_0.2deg_PT3H-i***", which begins in April 2023. The temporal extension of the original WAVERYS dataset therefore spans January-February-March 2023.

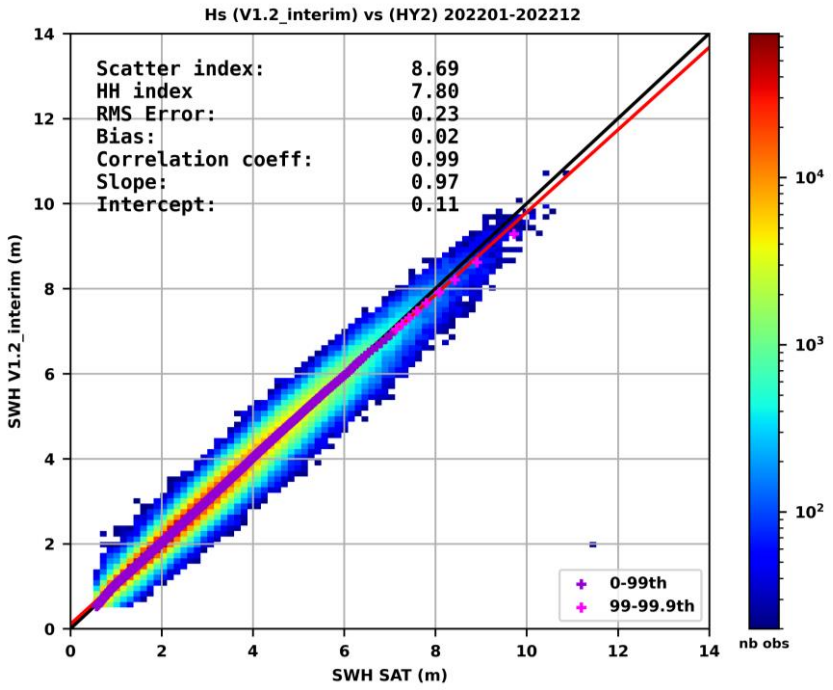


Figure 34. Top panel (this page): Scatter diagram between the SWH measured with HY-2B and C and WAVERYS for the 2022 time extension. The colour shade represents the observation density. Statistical scores are given, as well as the slope and y-axis of the associated linear regression. The crosses represent quantiles vs. quantiles values, the pink ones being for the highest values (99-99.9th percentiles). Bottom panel (next page): variation of statistical scores with latitude (continues in next page).

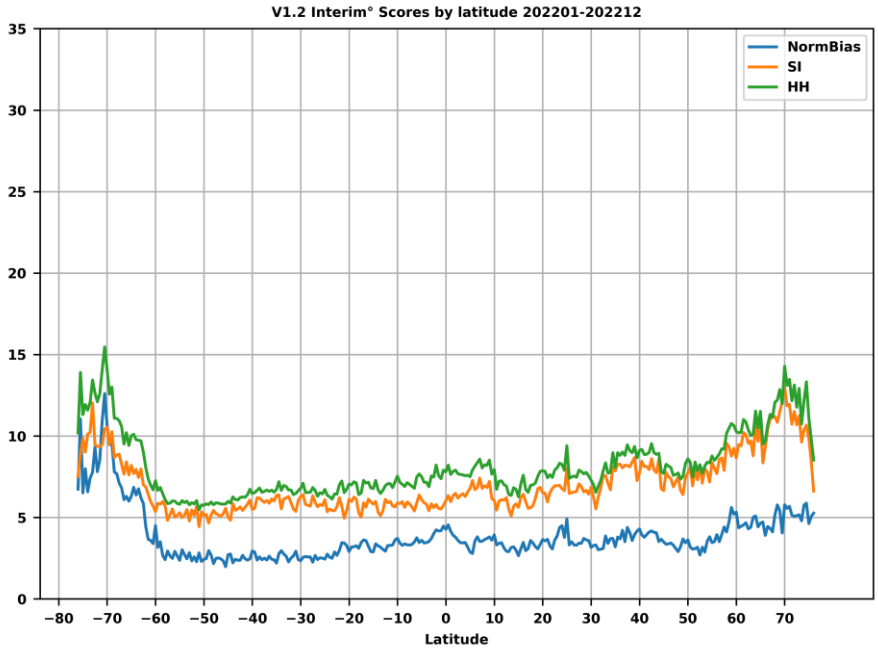


Figure 34 (cont.). Top pane (previous page)l: Scatter diagram between the SWH measured with HY-2B and C and WAVERYS for the 2022 time extension. The colour shade represents the observation density. Statistical scores are given, as well as the slope and y-axis of the associated linear regression. The crosses represent quantiles vs. quantiles values, the pink ones being for the highest values (99-99.9th percentiles). Bottom panel (this page): variation of statistical scores with latitude.

VI.3.2 Interim production

This update sets up “interim” production. Interims consists of a monthly production closer to real time than the current annual extension. Each M month, the data for the M-4 month are produced and delivered. Interims are equivalent in performance to real time extensions for WAVERYS, as four months are sufficient to optimally process all the data to be assimilated. For example, data for July 2023 will be available as Interims in November 2023. Under the previous system with annual extension, this data would only have been available in October 2024 (annual EIS of Copernicus Marine service), which represents a major improvement in terms of user availability. Interim production is delivered in the interim dataset "**cmems_mod_glo_wav_myint_0.2deg_PT3H-i**" and starts in April 2023.

VI.3.3 Climatology dataset

This update also sets up a monthly climatology dataset, computed from 1993 to 2020. It can be found under the dataset: “**cmems_mod_glo_wav_my_0.2deg-climatology_P1M-m**”. Only the VHM0 and VTM02 variables are distributed for this monthly climatology.

VI.4 Backward extension of time series: 1980 to 1992 (EIS November 2024)

The WAVERYS time series has been extended back in time to the 1980s. Production of this additional series was carried out as follows:

1/ production of a one-month spin-up on December, 1979

2/ run of WAVERYS V1.2 configuration without ocean currents and assimilation for January 1980 to January 1993. For this period, ocean currents are not applied, and the system is free from any assimilation of observed data due to the lack of GLORYS 12 currents and satellite missions.

3/ In order to provide a smooth transition between the two time series (the 1980-1992 unassimilated one and 1993-present assimilate one), January 1993 is reconstructed by combining the two series in a linear way, gradually bringing the assimilated solution into the output data. Noting $W_{pre-alti}$ the non-assimilated time series and W_{alti} the assimilated time series over January 1993:

$$W(t) = [1 - \alpha(t)]W_{pre-alti}(t) + \alpha(t)W_{alti}(t)$$

with $\alpha(t) = \frac{1}{31}$ t ; with t representing a day number of January 1993. The idea is to offer a smooth transition between $W_{pre-alti}$ and W_{alti} over the month of January 1993. To blend the two series, we propose a linear combination that gives more weight to the pre_alti period before January 15, 1993, then less afterwards, as the coefficients evolve with the date. For instance, for the date of January 10, α=0.32, so that $W \approx {}^{2}/_{3}\, W_{pre-alti} + {}^{1}/_{3}\, W_{pre-alti}$

As standard validation data such as buoys and altimeters are unavailable for this period, there is no other way than comparing WAVERYS to other historical series derived from models like ERA5 over the 80-90 period.

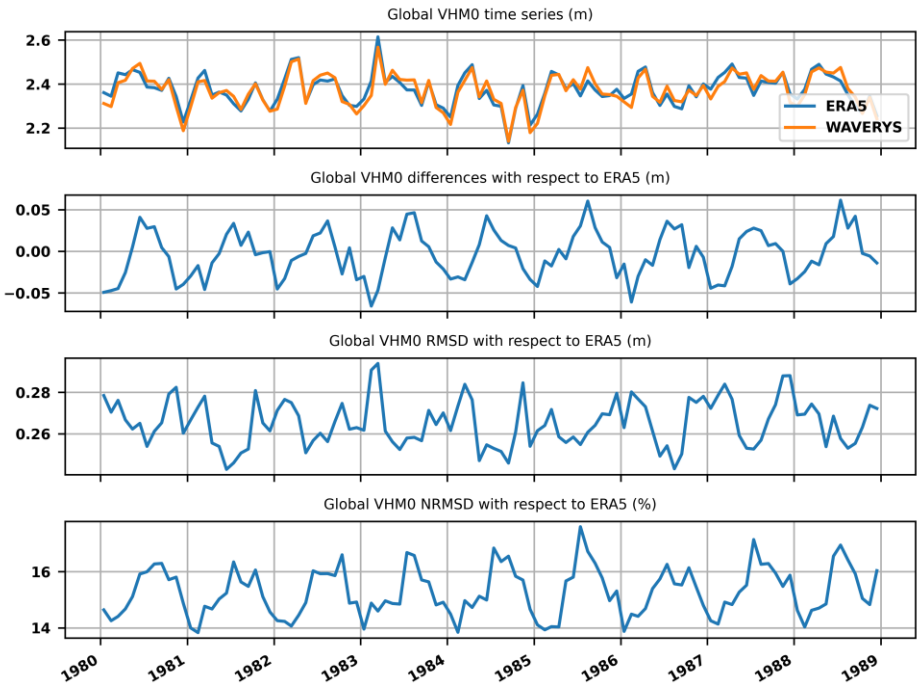


Figure 35. Comparison of the SWH time series of WAVERYS and ERA5 on the 1980-1989 period.

Figure 35 shows that the time series of the global mean SWH of WAVERYS and ERA5 are very similar over the historical period 1980-1989, which is expected since both wave systems use the same wind forcing. However, there are seasonal variations in the time series of the differences: WAVERYS is more energetic than ERA5 in austral winter and the opposite is true in austral summer, illustrating their differences in terms of physics and response to strong winds. WAVERYS is also closer to ERA5 in austral summer when global winds are weaker.

While it is not possible to better qualify the system for these historical years, the results for recent years encourage confidence in the quality of WAVERYS, which is equivalent to, if not better than, that of ERA5 over the period 1980-1993.

VI.4.1 List of bug fixes

VI.4.1.1 Unsorted latitude

The previous WAVERYS time series featured a grid with latitudes arranged in descending order. This could lead to difficulties when using certain software programs. Latitudes are now arranged in ascending order to facilitate manipulation of WAVERYS data.

VI.4.1.2 Bias correction for 2020 onwards: Sentinel1/CFOSAT SAR data assimilation + wave propagation bias.

Two anomalies have been detected and corrected for the 2020-2021 period. These anomalies concerns:

- Incorrect compilation when generating the propagation grids used by the wave model due to a change in the computing machine (compilation options were not equivalent between the two machines and produced different results). These anomalies produce large-scale negative biases of about 20 cm, especially in storm tracks, as illustrated in Figure 36.

NB: We encourage those who have carried out studies on extreme events during this period to re-upload the data and update their results.

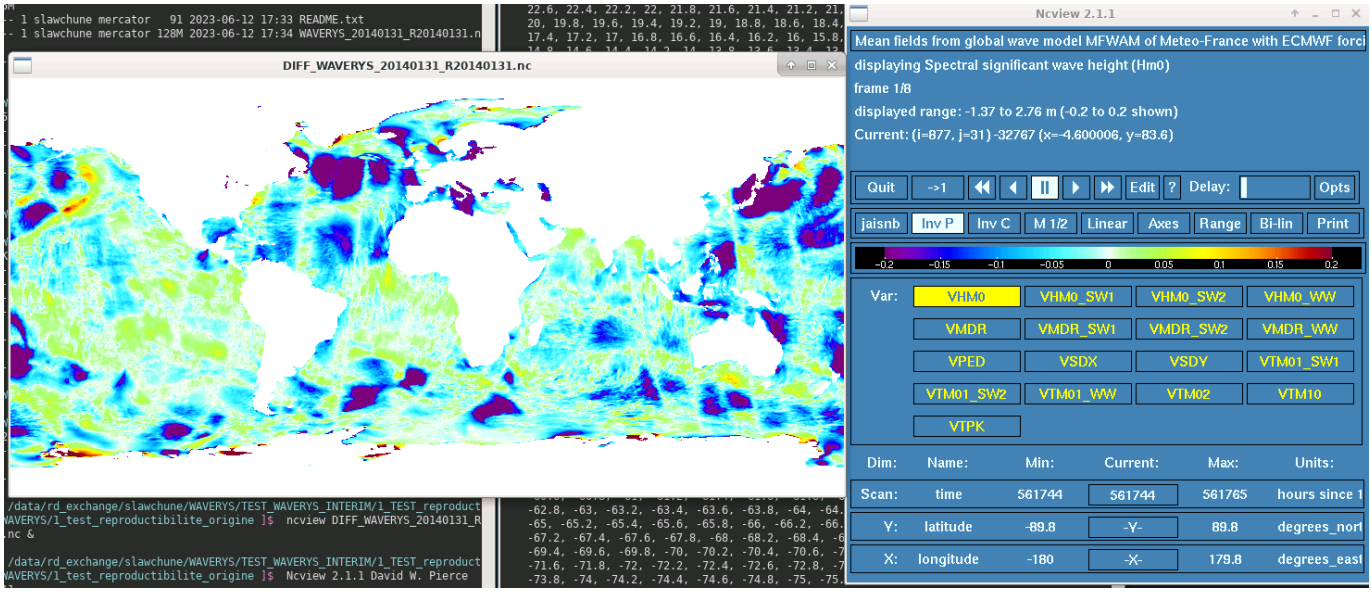


Figure 36. Impact of incorrect compilation of propagation grids on SWH (instantaneous value for 31st January 2014 at 00:00).

- Incomplete SAR data assimilation due to a data retrieval issue (archiving machine failure) during code execution. The impact in terms of instantaneous significant wave height are differences in wave height signatures that appear at the level of satellite tracks and can exceed 20 cm in magnitude. This is documented in Figure 36.

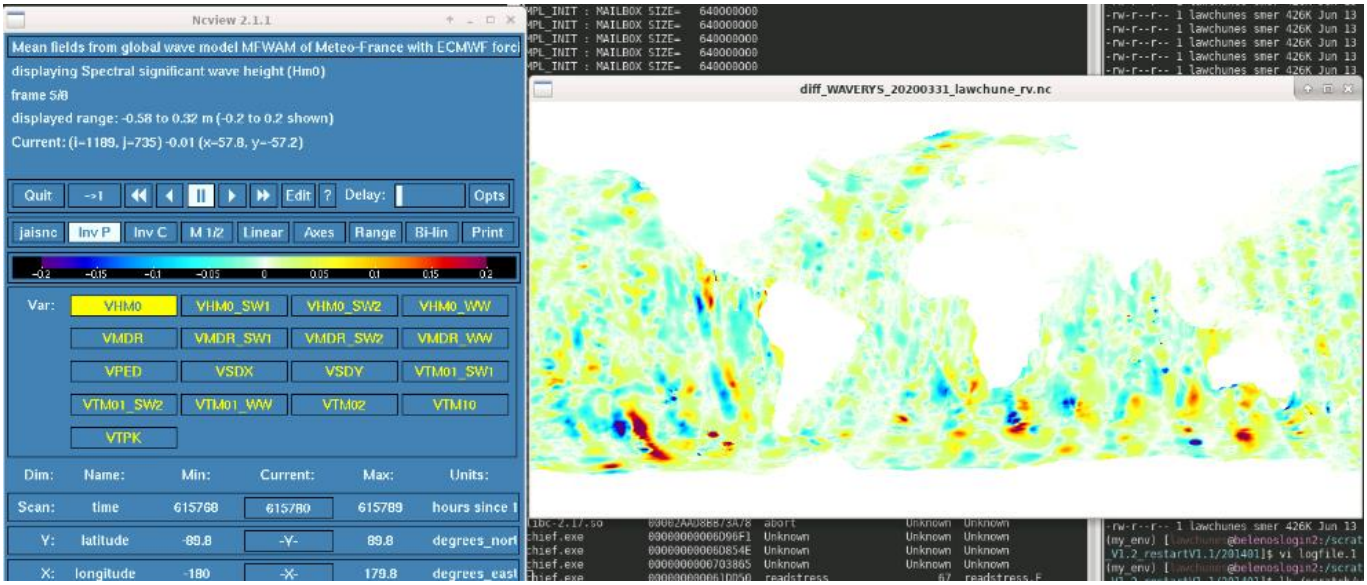


Figure 37. Impact of lack of SWIM/CFOSAT data on SWH (instantaneous value for 31st January 2014 at 00:00).

Figure 38 shows the impact of the bug correction on January 2021, by validating the significant height against the HY2b independent altimeter. Bug correction reduces the average mid-latitude bias (now, -3.5 cm vs. -5.9 cm previously) and improves statistical scores everywhere except at the equator.

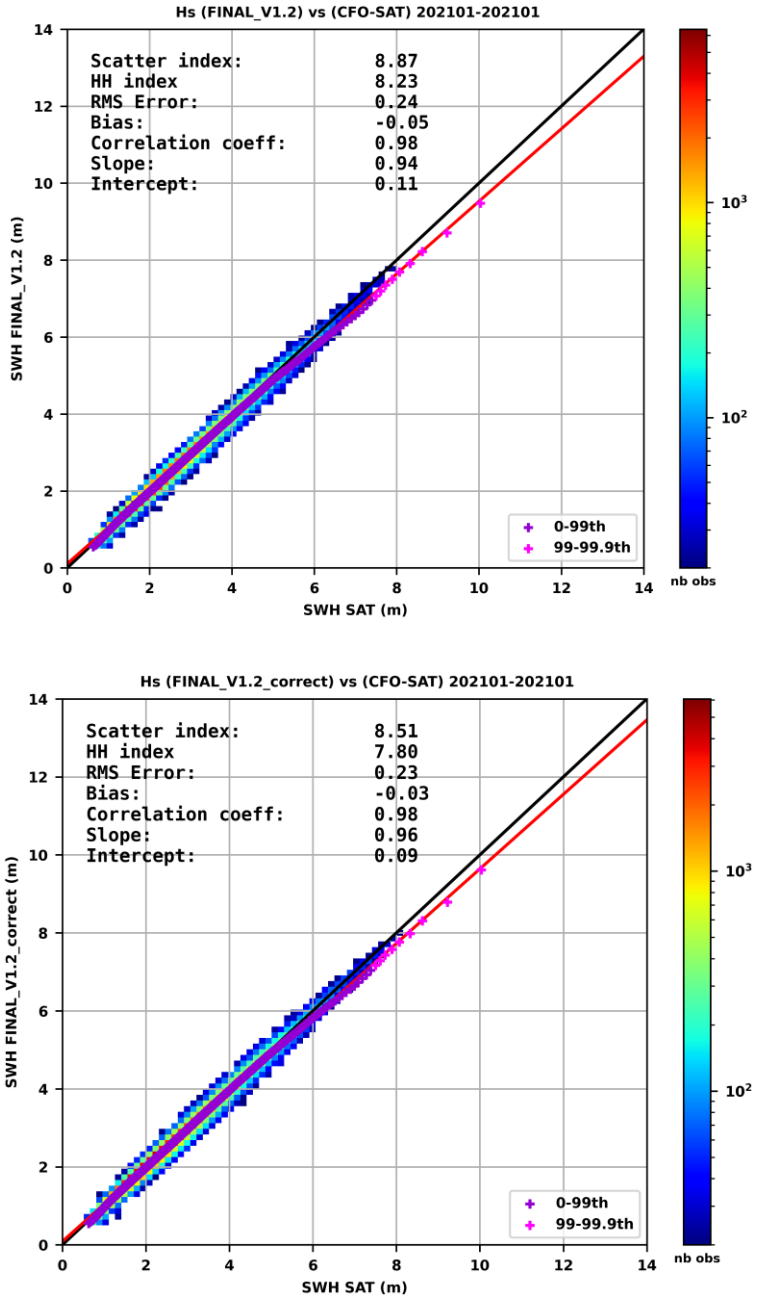


Figure 38: Scatterplot between HY2-b and models for the original V1.2 time series and the corrected one (this issue), for the month of January 2021.

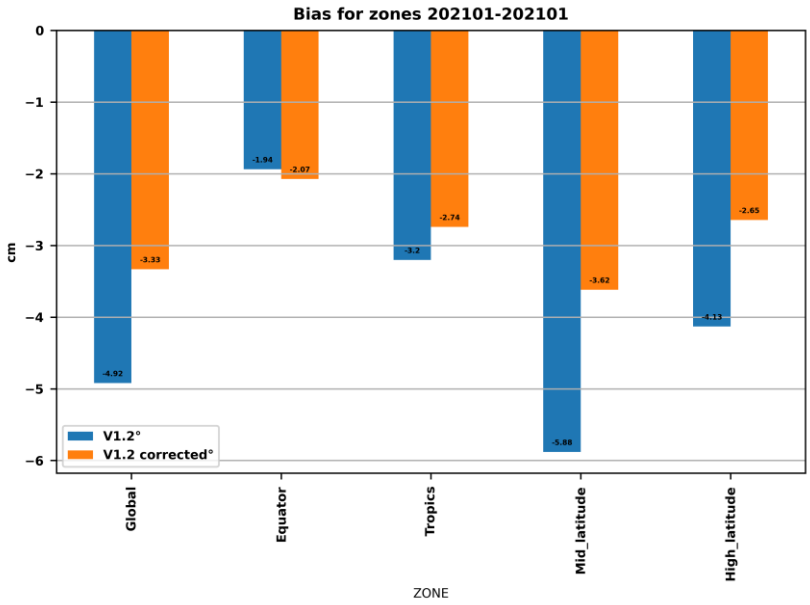


Figure 39: SWH bias for several latitudinal bands between the V1.2 and V1.2 corrected versions. Supporting observation dataset are HY2-b altimeter.

VI.4.1.3 New metadata updates in files

Metadata referring to the current dataset and product name new nomenclature have been updated for the entire series (for instance GLOBAL_MULTIYEAR_WAV_001_0032 instead of GLOBAL_REANALYSE_WAV_001_0032).

VII APPENDIX

VII.1 Appendix 1: Impact of ERA5 and ERA Interim for wave forecast.

Figure 40 illustrates the impact study conducted by Météo-France on sea state forecasts driven by different atmospheric reanalysis from ECMWF, namely: IFS operational winds, ERA5 winds, and ERA-Interim winds. The case study chosen is Hurricane Matthew in the Caribbean Sea. The hurricane's core is easily identified for wave forecasts forced by the IFS operational winds (left panel) with significant wave height of about 7 meters Northeast of Martinique Island. The results using ERA5 winds show similar features, although of slightly smaller SWH, reaching approximately 5 to 6 meters near the eye of the hurricane. Northeast of the domain the SWH from ERA5 is 3.5 meters which is smaller in this area than the results from IFS with 4.5 m. The ERA Interim winds degrade significantly the wave forecast in terms of front description and propagation. Quasi-spherical wave trains are shown to travel from the centre of the hurricane when using ERA Interim forcing, this results in poor resolved phase results because of the 6-hourly wind forcing. ERA interim misplaces completely the wave features observed by the wave forecast using ERA5 and IFS forcing.

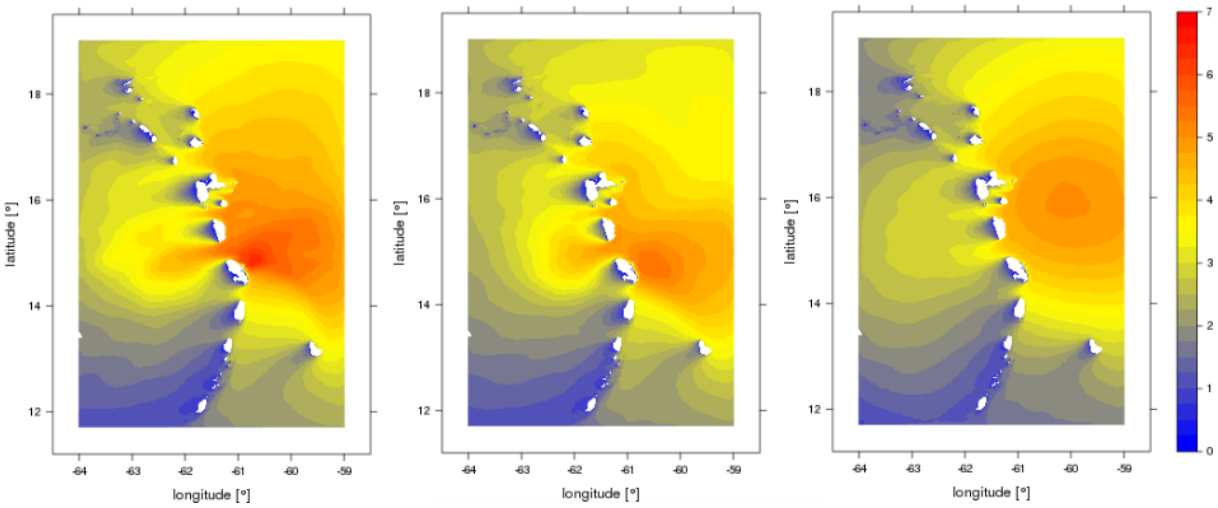


Figure 40.Significant wave height fields in meters from MFWAM model runs with different wind forcing in the Caribbean Sea during Hurricane Matthew on 28 September 2016 at 19:00 UTC. The wave model used is a regional configuration from Météo-France (WaveWatch3 with unstructured grid) nested in the global operational wave system MFWAM. From left to right, SWH forced by: IFS operational winds, 1 h and 30 km ERA5 winds, 6 h and 70 km ERAI winds.

I.1 Appendix 2: Sensitivity studies on the impact of forcing components.

In preparation for the design of WAVERYS, several test runs were carried out to investigate the best forcing frequency for the reanalysis. These tests are performed for full year of 2010 and significant wave heights observed by Jason2 were not included in the assimilation and were used as independent wave data for the validation. It resulted clearly that the 3-hourly winds induce the best performance on SWH. We also showed that the use of ERA5 significantly improves the scatter index of SWH in comparison with the run forced with ERA interim, as illustrated in Figure 41. The use of surface currents from GLORYS enhances the improvement for the scatter index of SWH mostly in large current systems for instance

the Gulf Stream Ocean area where the bias of SWH is significantly reduced. The assimilation of only two altimeter missions on 2010 (Envisat and Jason1) induces a significant correction on the mean wave parameters provided by the test run. The bias of SWH is then strongly reduced to globally 1 cm and 6 cm for the Southern Ocean. This study also provides an overview of the system's performance in reproducing a seasonal cycle (Figure 41, bottom). We observe a strong reduction of SWH bias from 25 cm to less than 10 cm when using the assimilation.

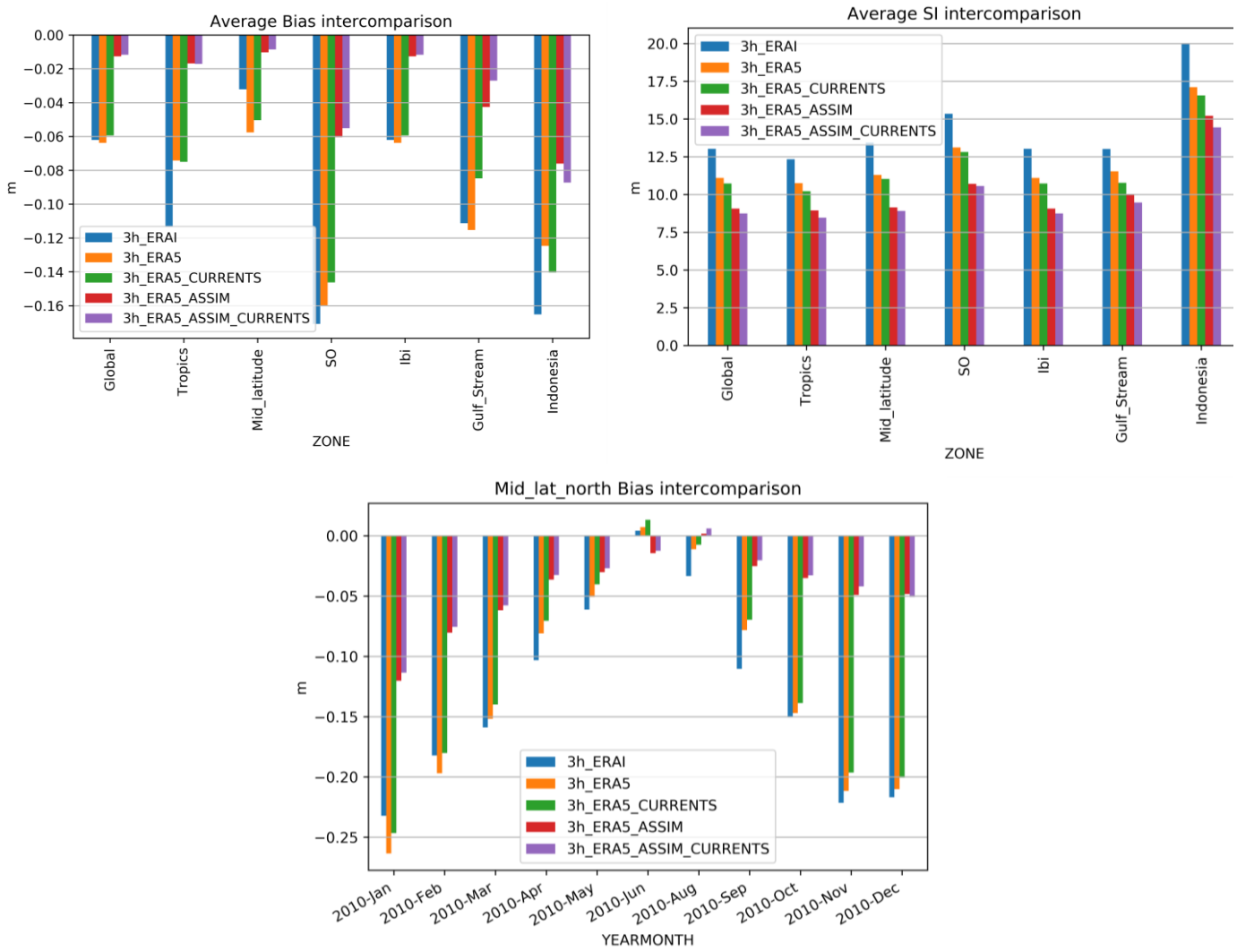


Figure 41. Performance of sensitivity tests of WAVERYS in 2010. The two top panels describe respectively the bias and SI of SWH for different ocean areas. Bottom panel stands for the monthly variation of SWH bias mid-latitudes of the northern hemisphere.

I.2 Appendix 3: Impact of large and erroneous ERA5 10 m winds in the Black Sea and Papua-Guinea

The atmospheric forcing from ERA5 has showed in some dates and locations abnormally high wind values for a few hours. These events seem to be related to numerical 'explosions' of the ERA5 model that produce unrealistically high values. The list of these events is available on the ECMWF website: ERA5: large 10m winds - Copernicus Knowledge Base - ECMWF Confluence Wiki

As a precaution in WAVERYS, winds are set at a maximum of 50 m/s (185 km/h). This section studies the impact of those unrealistic ERA5 wind bursts on the WAVERYS analyses. Two grid points are systematically involved for WAVERYS. They are situated on the coasts of the **Black Sea** and **Papua-Guinea**. provides information on the dates of these episodes as well as the maximum value of obtained Hs. Fortunately, the unrealistically generated wave trains in the range of 3-8 m (in WAVERYS) quickly dissipate most of the time (timescales of the order of a day) due to the short duration of the wind pulse (i.e., short fetch). For a few of these episodes, however, these unrealistic wave trains can slightly propagate in time, as illustrated in Figure 42 and Figure 43. These wave trains can flow along the coastline of the Arafura Sea (for Papua New Guinea), and flow as a swell from the cities of Rize to Sochi (for the Black Sea). **We invite users to be cautious about the degraded quality of WAVERYS for these cases**, especially the underlined cases in and

. There are other frequent explosion sites for ERA5 winds such as Antarctica, China, Ecuador and Kyrgyzstan, but these do not have an impact on WAVERYS' computing domain.

Table 9. Date of unrealistic wind pulse events and maximum significant wave height generated in the Black Sea. An underlined date indicates a significant propagation of the generated wave trains (not dissipated locally).

Site : Black Sea, [40.89°N, 39.6°E]	
Time	Maximum SWH (m)
1995-01-27 21:00	4.3
1997-04-22 21:00	5.15
2001-01-27 09:00	6.72
2005-12-08 09:00	3.82
2006-12-10 09:00	2.5
2010-01-10 21:00	4.04
2011-12-08 00:00	4.4
2013-12-24 09:00	5.83

Table 10. Date of unrealistic wind pulse events and maximum wave height generated in the Papua-Guinea. An underlined date indicates a significant propagation of the generated wave trains (not dissipated locally).

Site : Papua-Guinea, [5.199°S, 137.531°E]	
Time	Hs max generated value (m)
1996-07-10 21:00	5.27
1997-06-27 00:00	4.41
1997-08-18 00:00	4.72
2003-05-17 00:00	3.85
2004-08-04 00:00	4.04
2004-08-31 00:00	7.02
2005-08-06 00:00	5.48
2006-06-12 21:00	6.98
2012-06-29 21:00	4.18
2015-09-12 21:00	3.29
2016-05-16 21:00	6.21
2016-05-31 21:00	4.72
2016-08-25 21:00	5.53
2017-10-11 21:00	4.58
2018-09-14 21:00	4.29

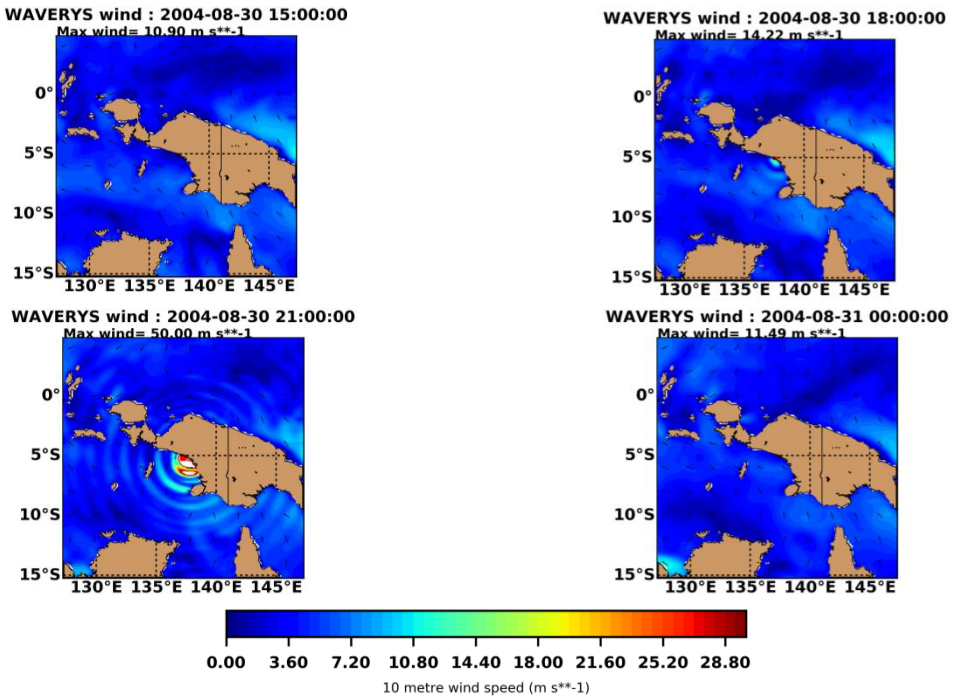


Figure 42. 10 m wind chronology from ERA5 wind forcing dataset for a particular spurious event in the Guinean Sea. Maximum wind was higher than 50 m/s.

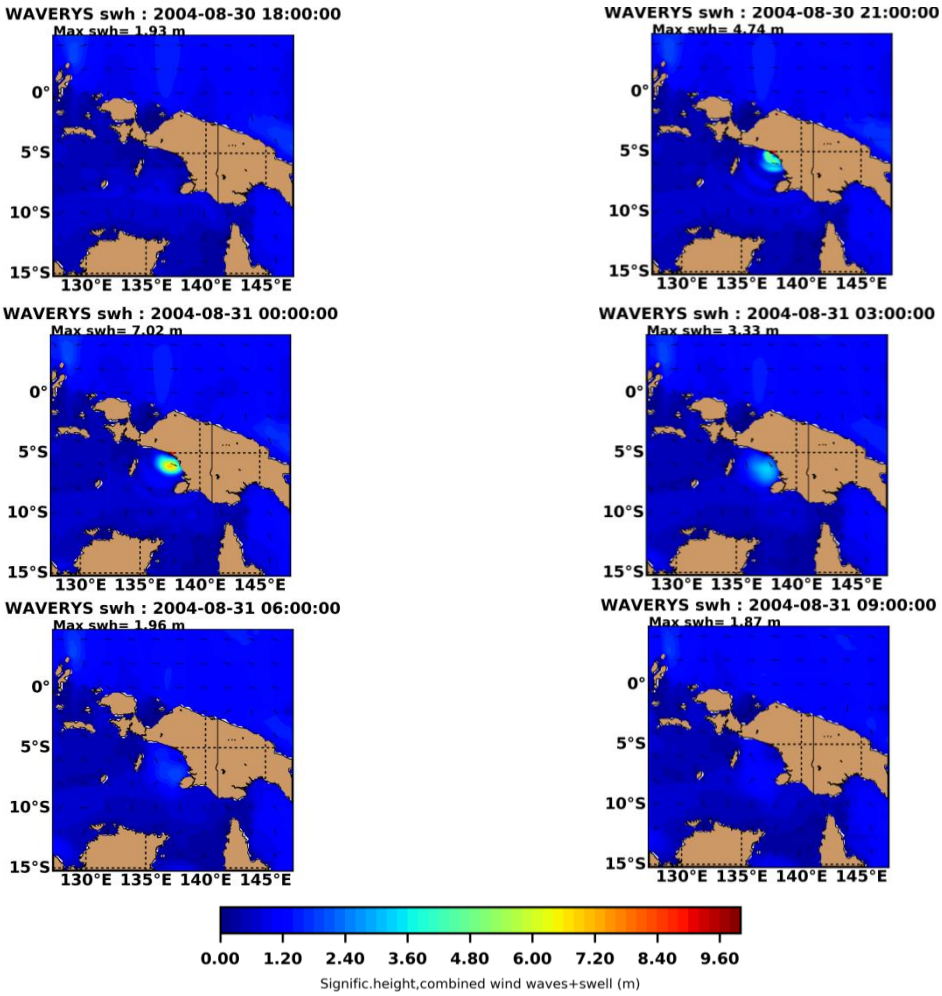


Figure 43. Illustration of wind-sea / swell generation to spurious wind event shown in Figure 42. Spurious waves are dissipated within a day.

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